IT3021

Data Warehousing & Business Intelligence

Assignment 2

Submitted By: Karunartahne U.M.T.N

IT23679726

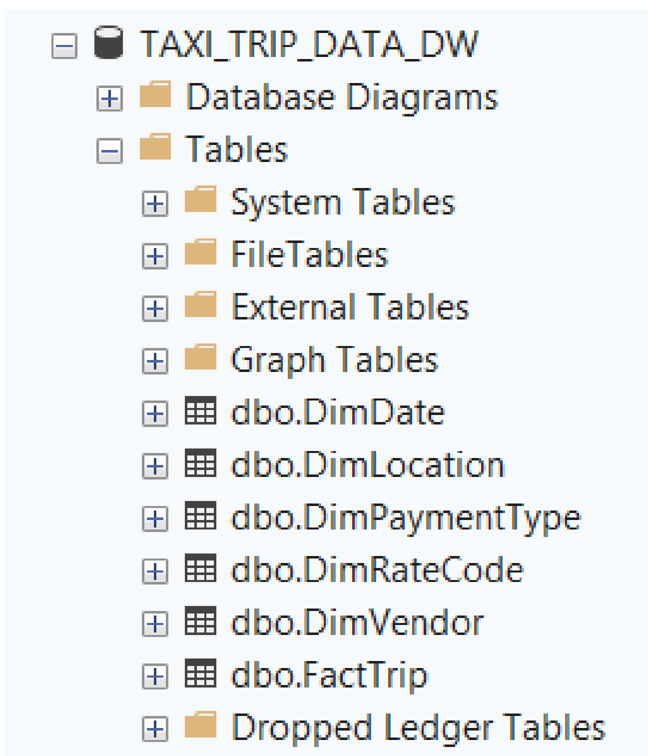
Table of Contents

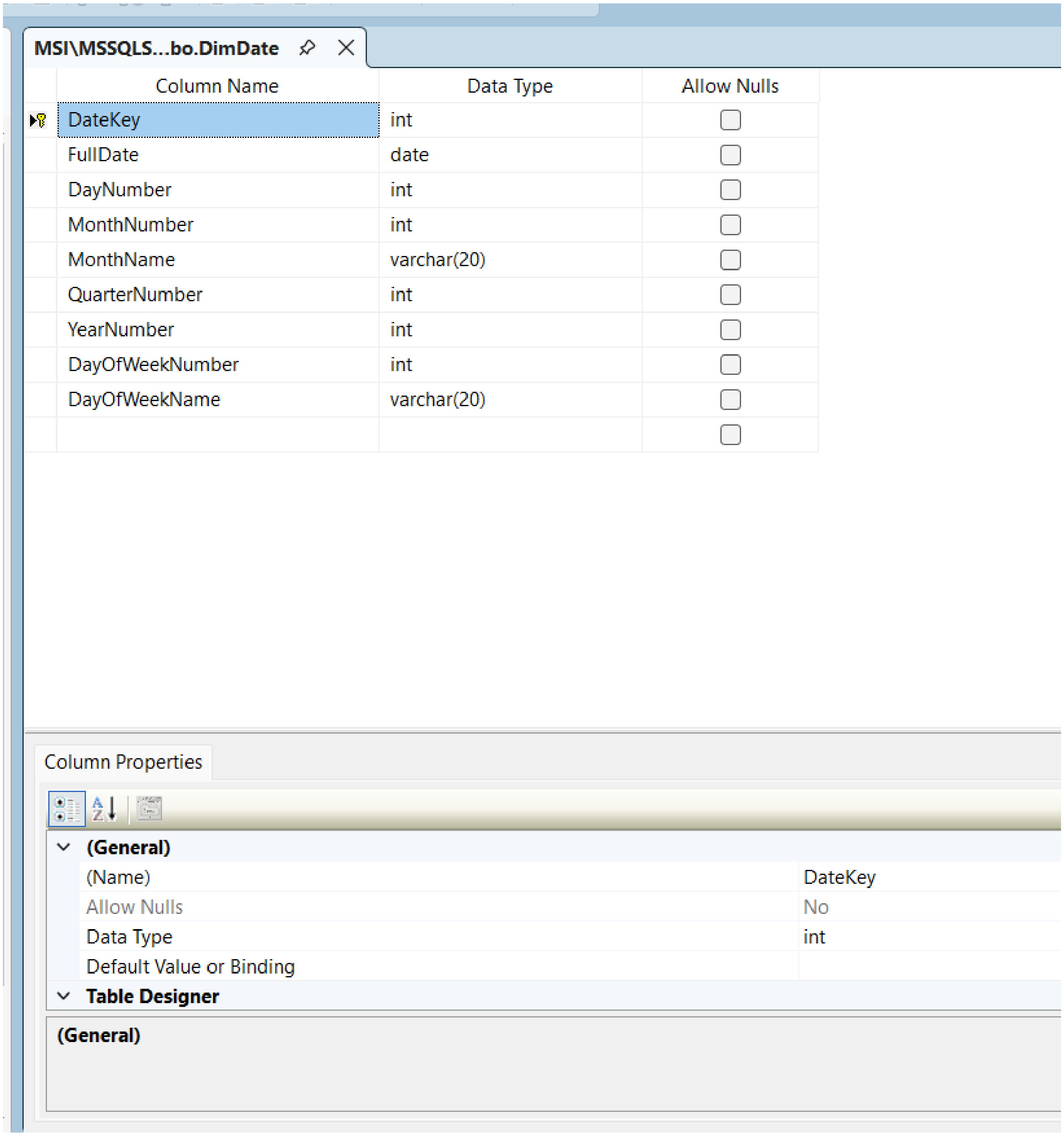
1. Data source for the assignment.

This assignment utilizes the data warehouse developed in Assignment 1 as the primary data source. The data warehouse is designed based on the NYC Taxi Trip dataset and follows a star schema architecture to support analytical processing and reporting.

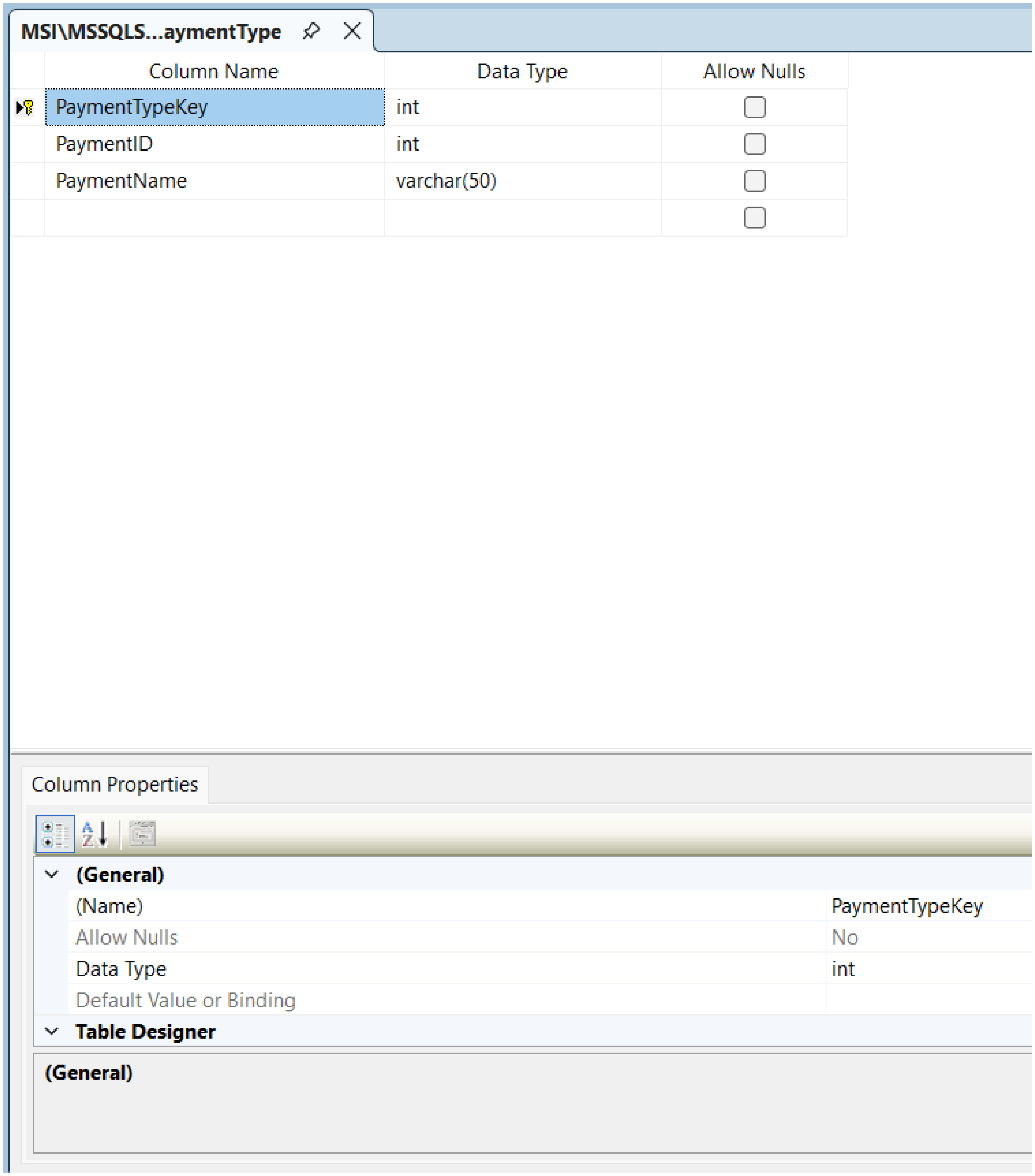
The central fact table in the warehouse is the FactTrip table, which stores transactional data for each taxi trip. Each record in the FactTrip table represents a single taxi trip and contains measurable attributes such as trip distance, fare amount, total amount, passenger count, and trip duration.

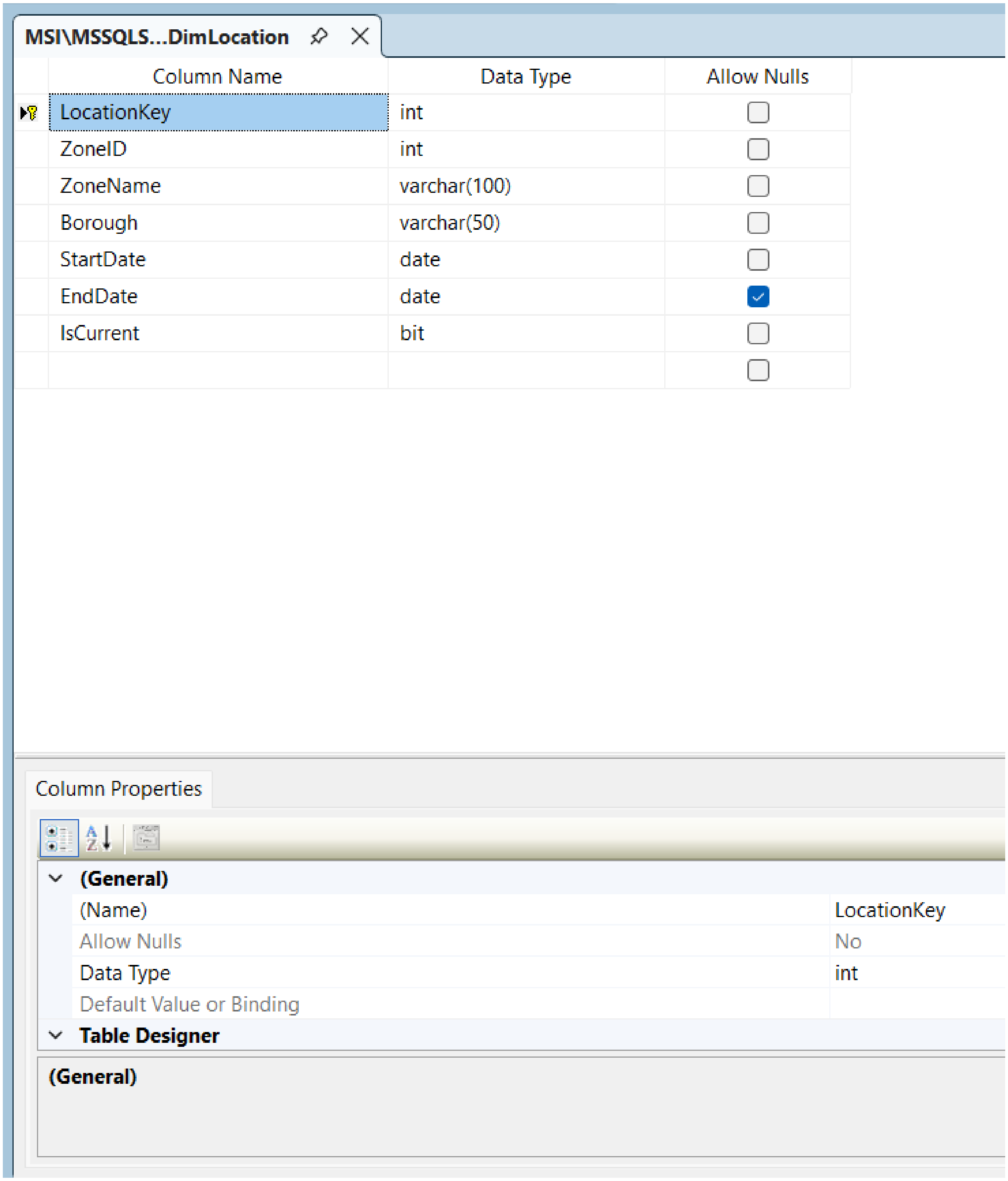
The data warehouse also implements Slowly Changing Dimensions (SCD), particularly in the DimLocation table, to track changes in location-related attributes over time.

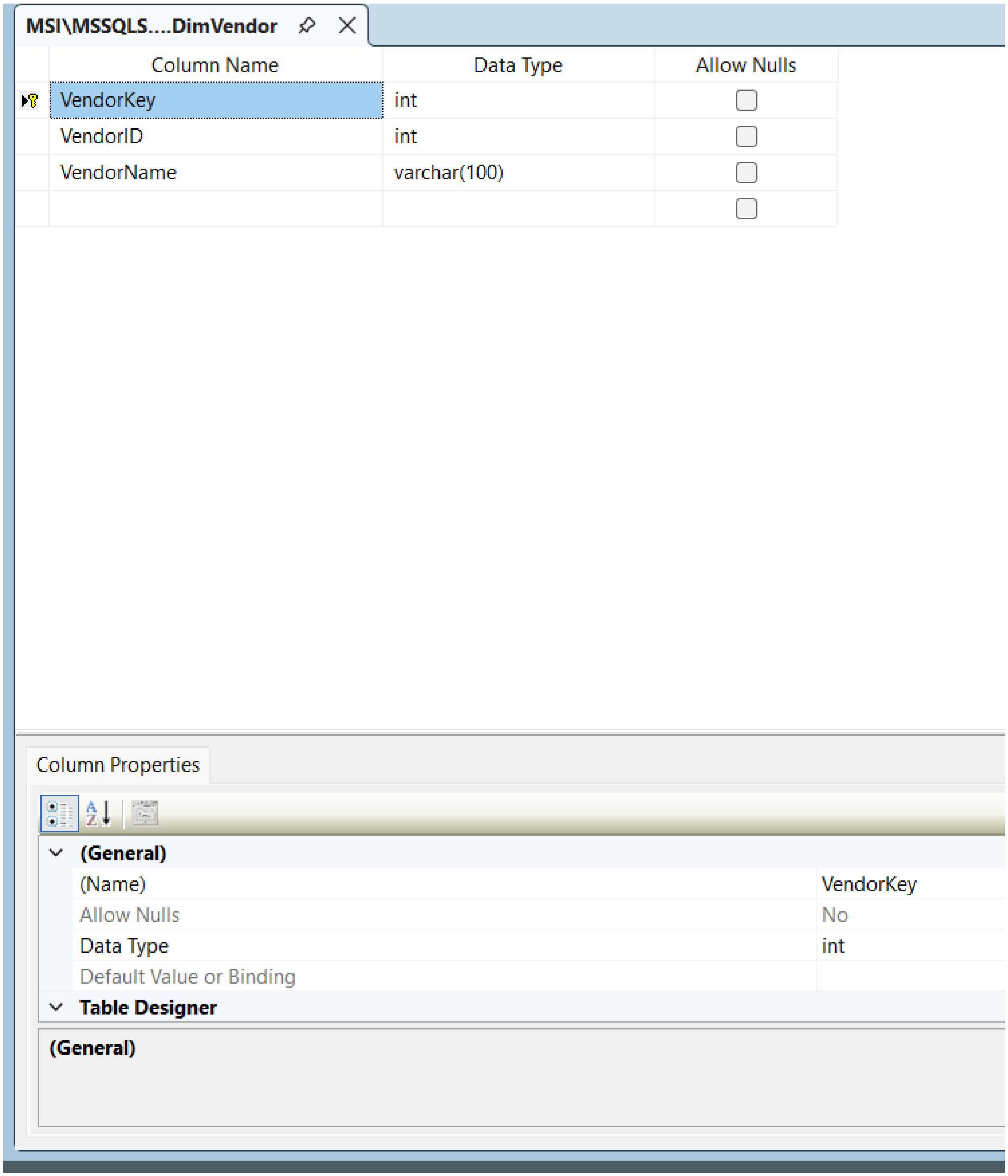


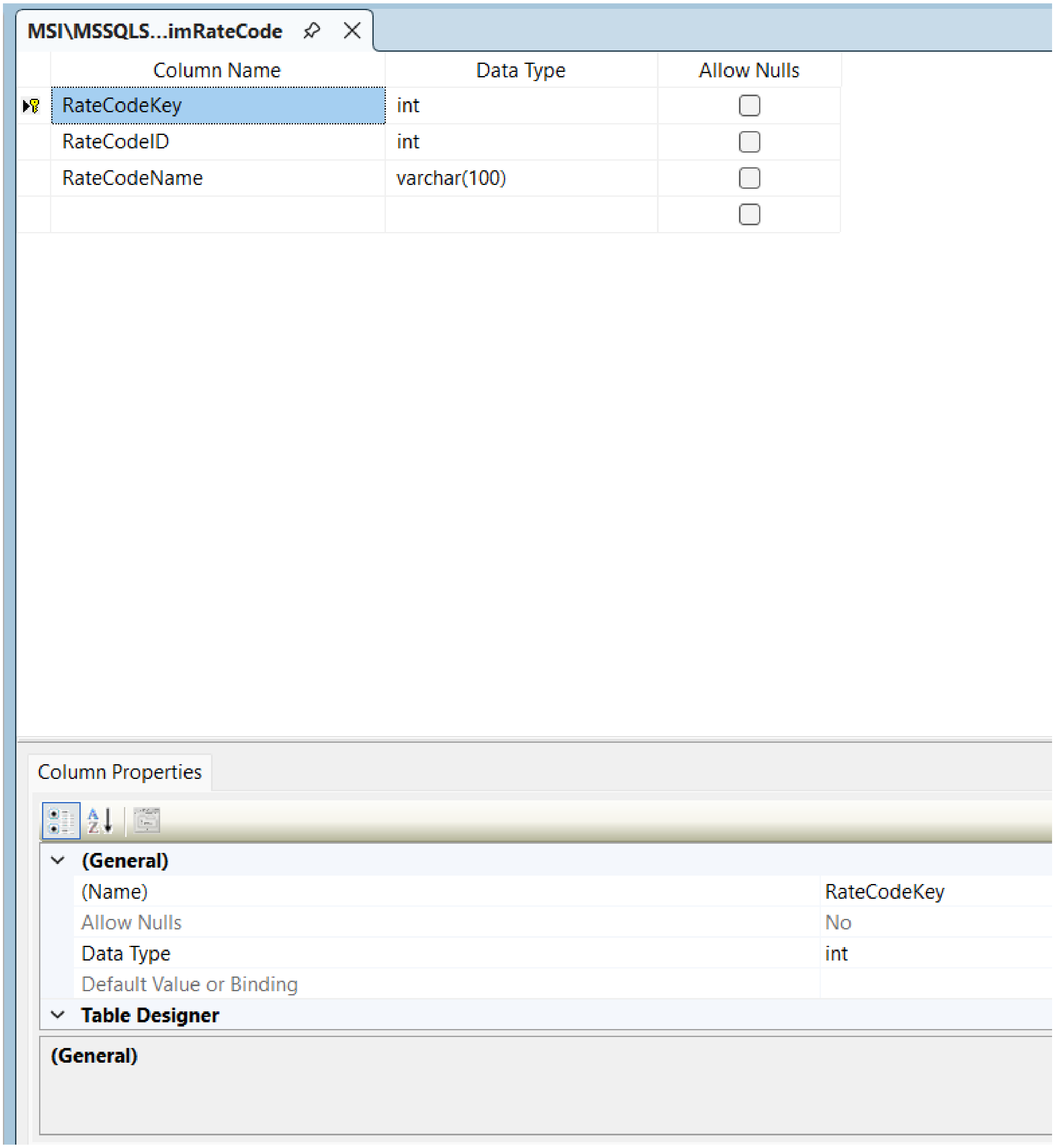


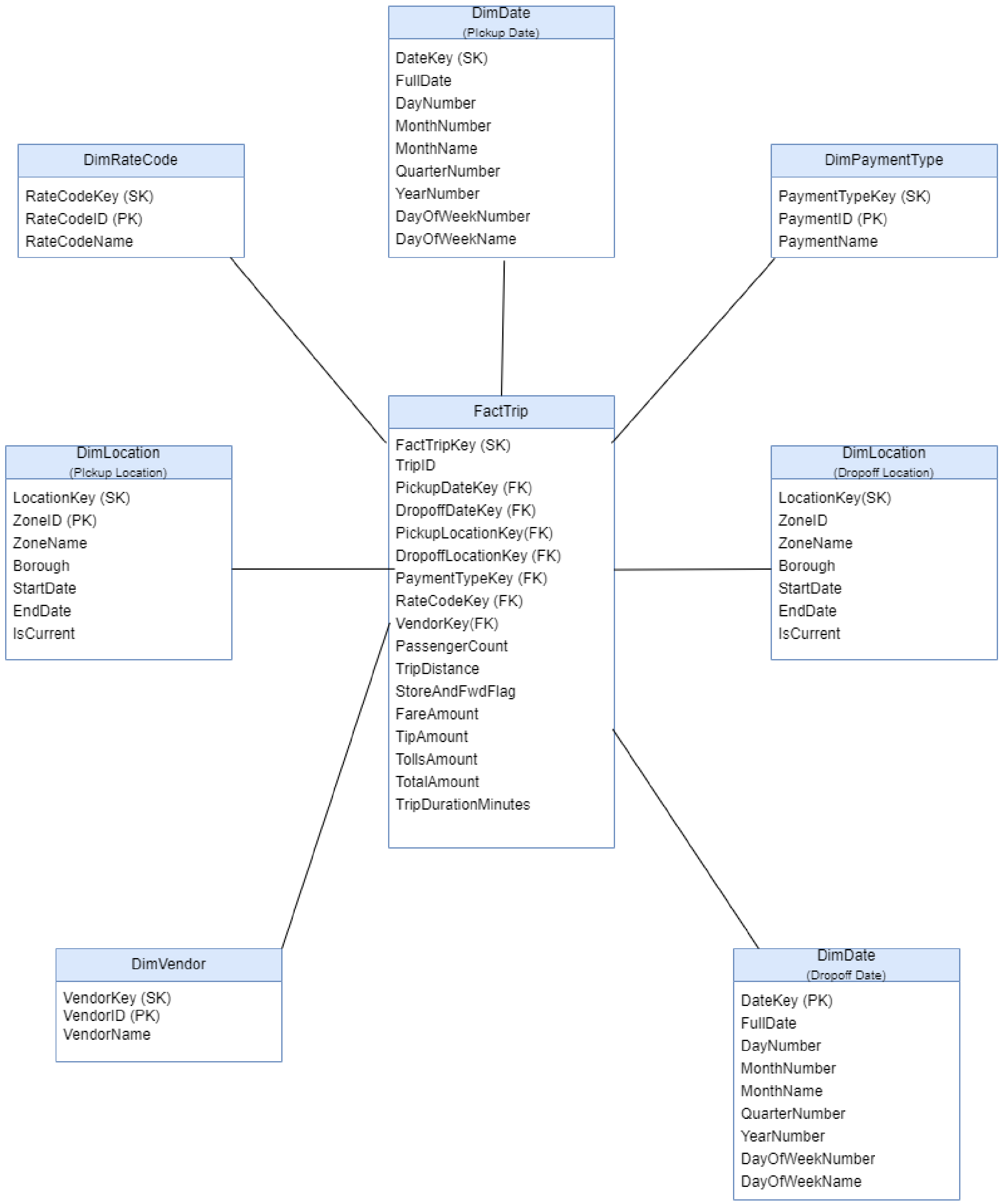
Column Name	Data Type	Allow Nulls
DateKey	int	
FullDate	date	
DayNumber	int	
MonthNumber	int	
MonthName	varchar(20)	
QuarterNumber	int	
YearNumber	int	
DayOfWeekNumber	int	
DayOfWeekName	varchar(20)	

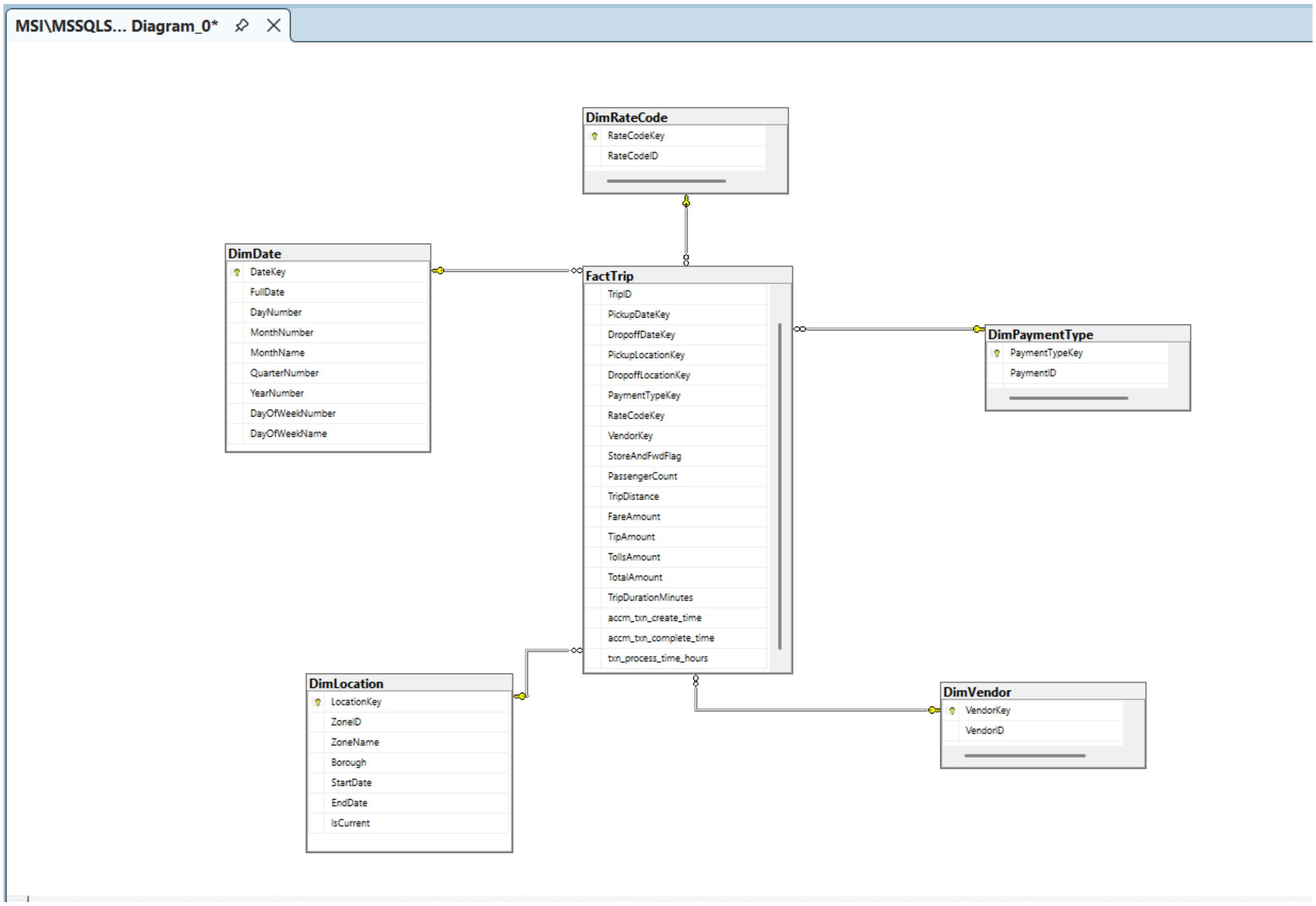

MSI\MSSQLS...aymentType
Column Name
Data Type
Allow Nulls
PaymentTypeKey
int
PaymentID
int
PaymentName
varchar(50)
Column Properties
(General)
(Name)
PaymentTypeKey
Allow Nulls
No
Data Type
int
Default Value or Binding
Table Designer
(General)


MSI\MSSQLS...DimLocation
Column Name
Data Type
Allow Nulls
LocationKey
int
ZoneID
int
ZoneName
varchar(100)
Borough
varchar(50)
StartDate
date
EndDate
date
IsCurrent
bit
Column Properties
(General)
(Name)
LocationKey
Allow Nulls
No
Data Type
int
Default Value or Binding
Table Designer
(General)


MSI\MSSQLS....DimVendor
Column Name
Data Type
Allow Nulls
VendorKey
int
VendorID
int
VendorName
varchar(100)
Column Properties
(General)
(Name)
VendorKey
Allow Nulls
No
Data Type
int
Default Value or Binding
Table Designer
(General)


MSI\MSSQLS...imRateCode
Column Name
Data Type
Allow Nulls
RateCodeKey
int
RateCodeID
int
RateCodeName
varchar(100)
Column Properties
(General)
(Name)
RateCodeKey
Allow Nulls
No
Data Type
int
Default Value or Binding
Table Designer
(General)


DimDate
(Pickup Date)
DateKey (SK)
FullDate
DayNumber
MonthNumber
MonthName
QuarterNumber
YearNumber
DayOfWeekNumber
DayOfWeekName
DimRateCode
RateCodeKey (SK)
RateCodeID (PK)
RateCodeName
DimPaymentType
PaymentTypeKey (SK)
PaymentID (PK)
PaymentName
FactTrip
FactTripKey (SK)
TripID
PickupDateKey (FK)
DropoffDateKey (FK)
PickupLocationKey(FK)
DropoffLocationKey (FK)
PaymentTypeKey (FK)
RateCodeKey (FK)
VendorKey(FK)
PassengerCount
TripDistance
StoreAndFwdFlag
FareAmount
TipAmount
TollsAmount
TotalAmount
TripDurationMinutes
DimLocation
(Pickup Location)
LocationKey (SK)
ZoneID (PK)
ZoneName
Borough
StartDate
EndDate
IsCurrent
DimLocation
(Dropoff Location)
LocationKey(SK)
ZoneID
ZoneName
Borough
StartDate
EndDate
IsCurrent
DimVendor
VendorKey (SK)
VendorID (PK)
VendorName
DimDate
(Dropoff Date)
DateKey (PK)
FullDate
DayNumber
MonthNumber
MonthName
QuarterNumber
YearNumber
DayOfWeekNumber
DayOfWeekName



100 % No issues found Ln: 11, Ch: 1 (34 chars, 1 lines) TABS CRLF UTF-8

Results Messages

	FactTripKey	TripID	PickupDateKey	DropoffDateKey	PickupLocationKey	DropoffLocationKey	PaymentTypeKey	RateCodeKey	VendorKey	StoreAndFwdFlag	PassengerCount	TripDistance	FareAmount	TipAmount	Tolls
1	1	TRIP_1	20180329	20180329	481	267	25	27	12	N	1	18.15	70.00	16.16	10.5
2	2	TRIP_2	20180329	20180329	443	494	25	25	12	N	1	4.59	25.00	5.16	0.00
3	3	TRIP_3	20180329	20180329	495	495	25	25	12	N	1	0.30	3.00	0.76	0.00
4	4	TRIP_4	20180329	20180329	495	349	25	25	12	N	2	16.97	49.50	5.61	5.76
5	5	TRIP_5	20180329	20180329	455	349	25	25	12	N	5	14.45	45.50	10.41	5.76
6	6	TRIP_6	20180329	20180329	451	349	25	25	12	N	1	11.60	42.00	14.57	5.76
7	7	TRIP_7	20180329	20180329	458	455	25	25	11	N	1	5.80	24.00	4.95	0.00
8	8	TRIP_8	20180329	20180329	475	481	25	25	12	N	1	3.38	25.00	5.16	0.00
9	9	TRIP_9	20180329	20180329	455	267	25	27	12	N	1	16.98	85.00	15.00	12.5
10	10	TRIP_10	20180329	20180329	443	481	25	25	12	N	1	4.99	22.00	4.76	0.00

100 % No issues found

Results | Messages

	LocationKey	ZoneID	ZoneName	Borough	StartDate	EndDate	IsCurrent
1	267	1	Newark Airport	EWR	2026-03-26	NULL	1
2	268	3	Allerton/Pelham Gardens	Bronx	2026-03-26	NULL	1
3	269	18	Bedford Park	Bronx	2026-03-26	NULL	1
4	270	20	Belmont	Bronx	2026-03-26	NULL	1
5	271	31	Bronx Park	Bronx	2026-03-26	NULL	1
6	272	32	Bronxdale	Bronx	2026-03-26	NULL	1
7	273	46	City Island	Bronx	2026-03-26	NULL	1
8	274	47	Claremont/Bathgate	Bronx	2026-03-26	NULL	1
9	275	51	Co-Op City	Bronx	2026-03-26	NULL	1
10	276	58	Country Club	Bronx	2026-03-26	NULL	1

	VendorKey	VendorID	VendorName
1	11	1	Creative Mobile Technologies, LLC
2	12	2	VeriFone Inc.

100 % No issues found

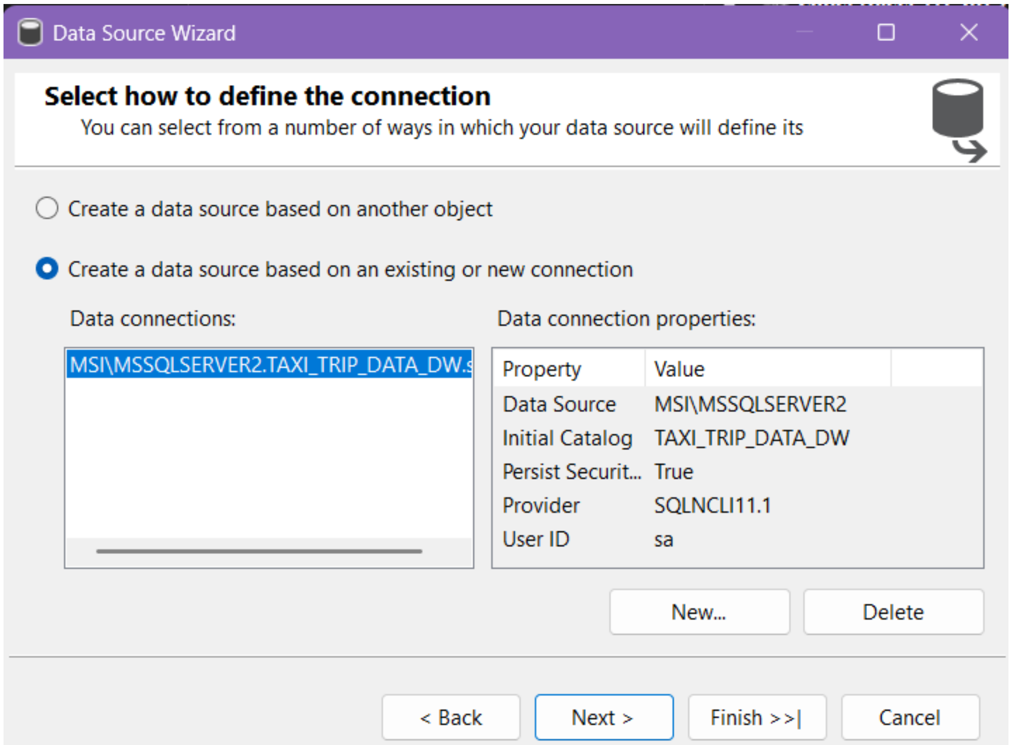
Results | Messages

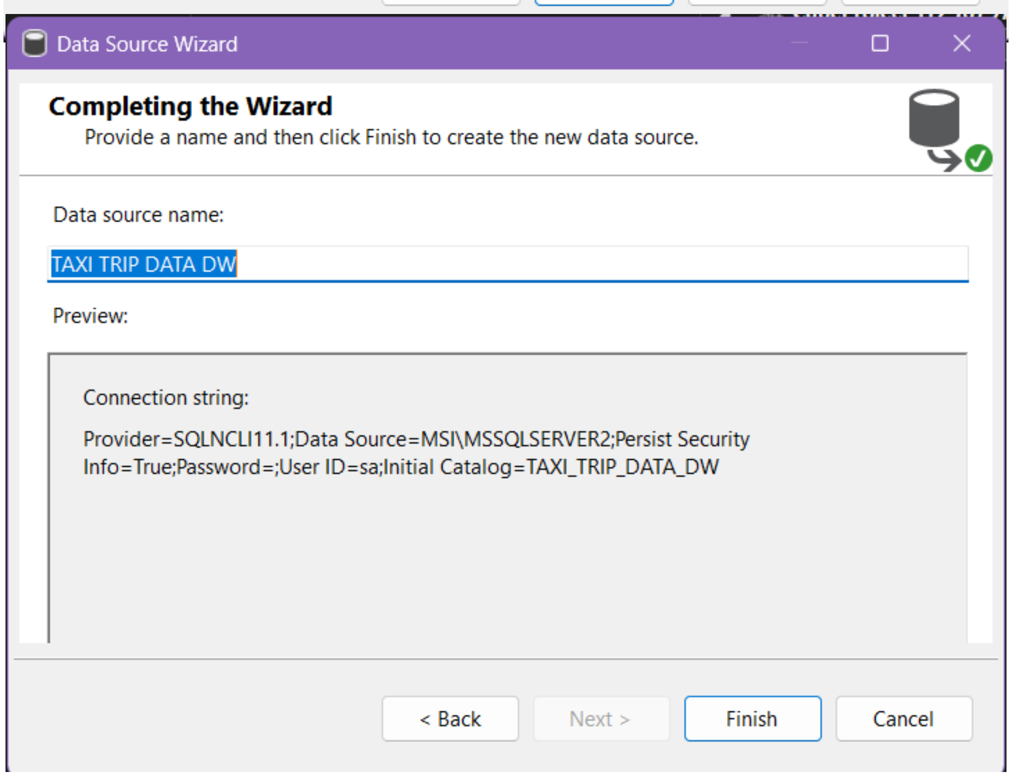
	PaymentTypeKey	PaymentID	PaymentName
1	25	1	Credit Card
2	26	2	Cash
3	27	3	No Charge
4	28	4	Dispute
5	29	5	Unknown
6	30	6	Voided Trip

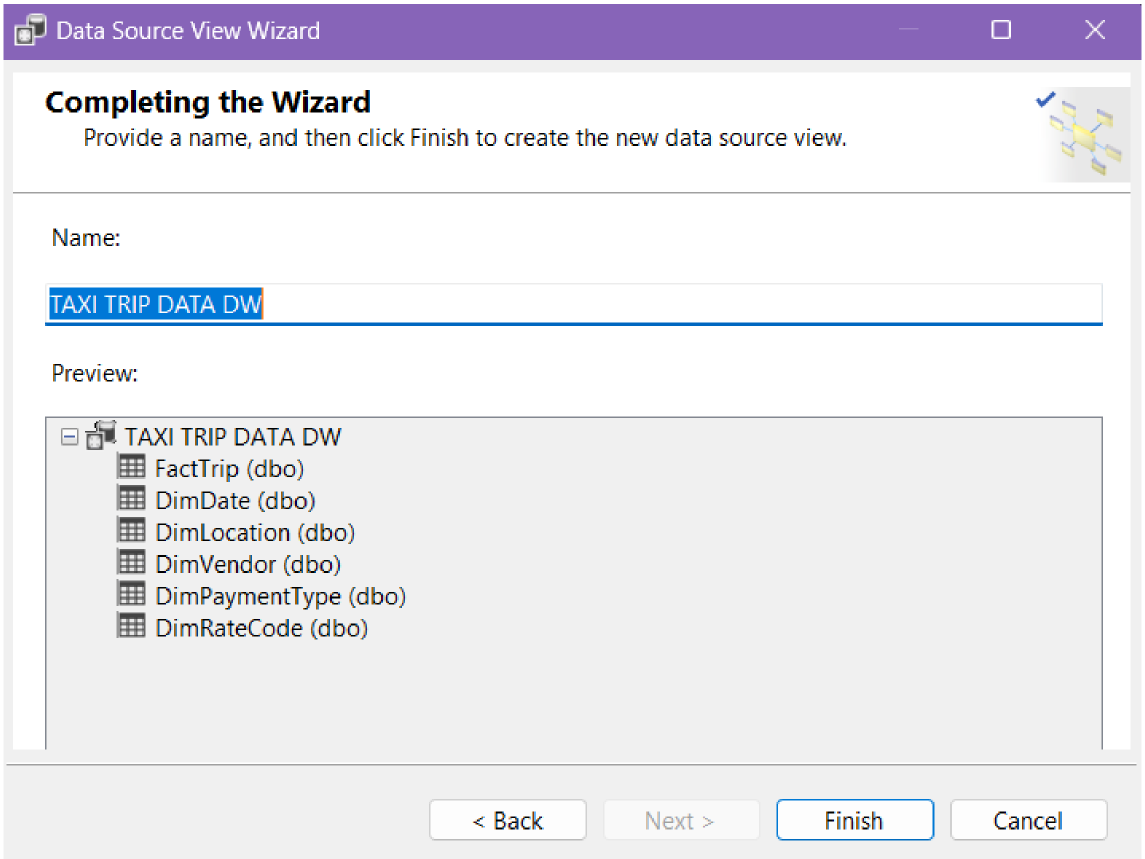
	RateCodeKey	RateCodeID	RateCodeName
1	25	1	Standard Rate
2	26	2	JFK
3	27	3	Newark
4	28	4	Nassau or Westchester
5	29	5	Negotiated Fare
6	30	6	Group Ride

2. <u>SSAS Cube implementation</u>

2.1.Create the Data Source

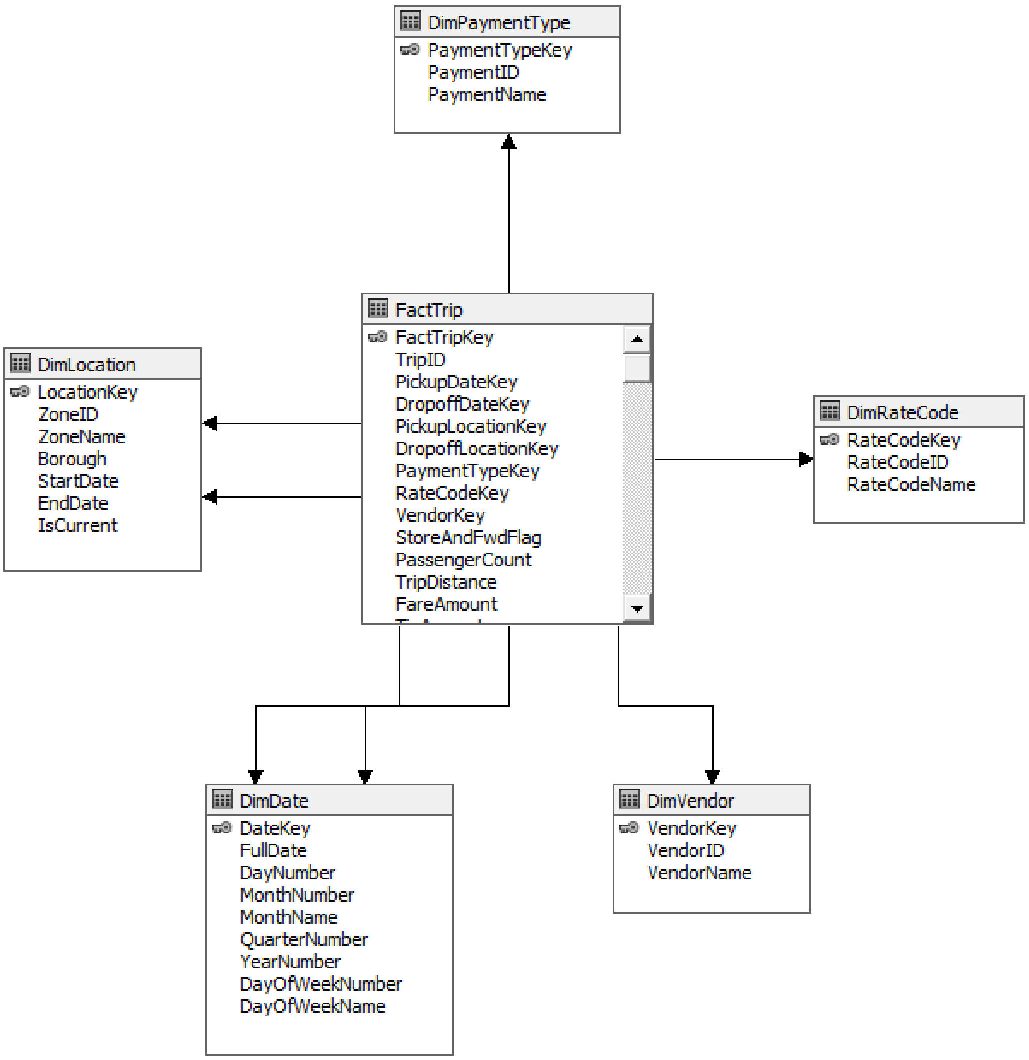




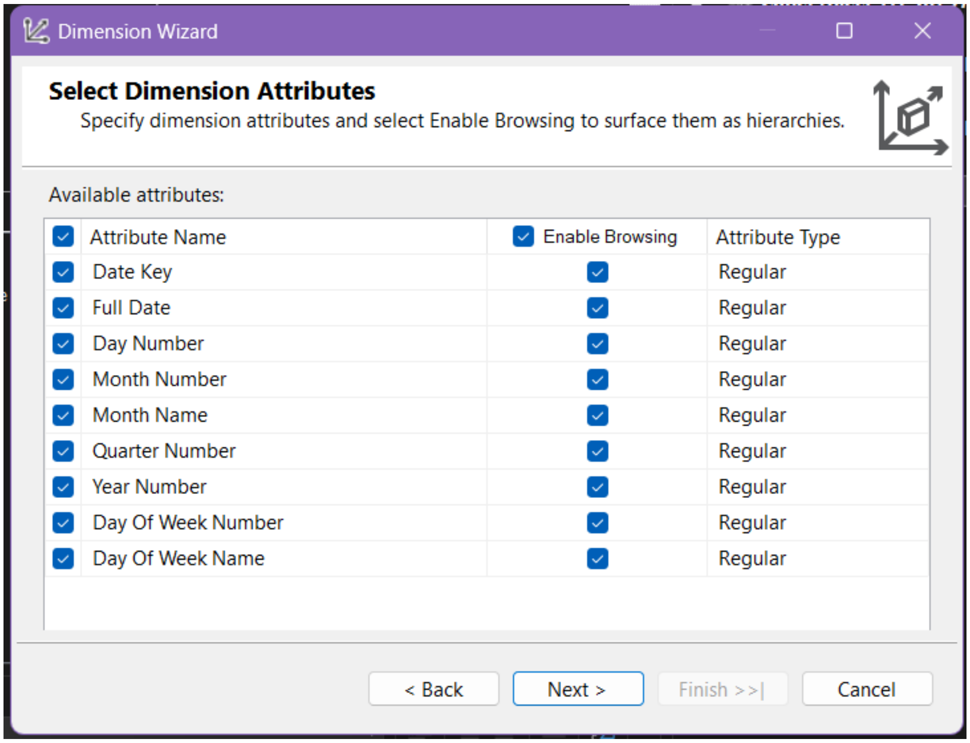


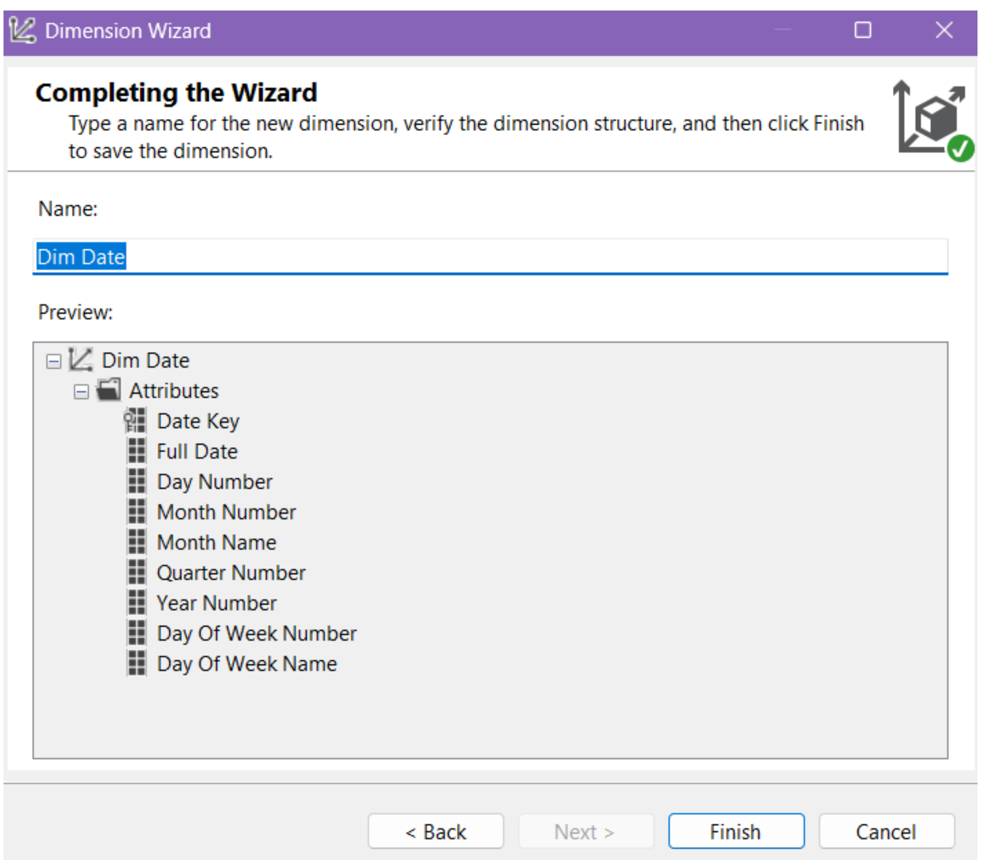
2.2.Create the Data Source View

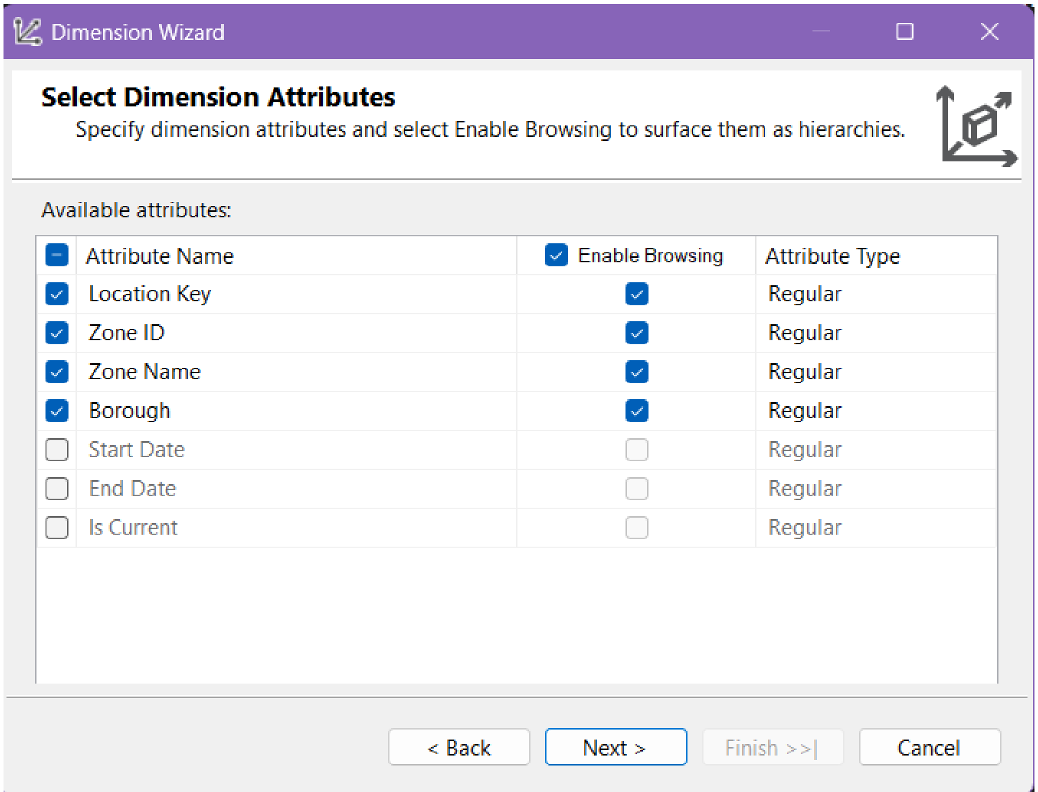
A Data Source View (DSV) was created to define the logical model used by the cube. The DSV displayed the relationships between the fact table and the dimension tables and served as the basis for dimension and cube creation.

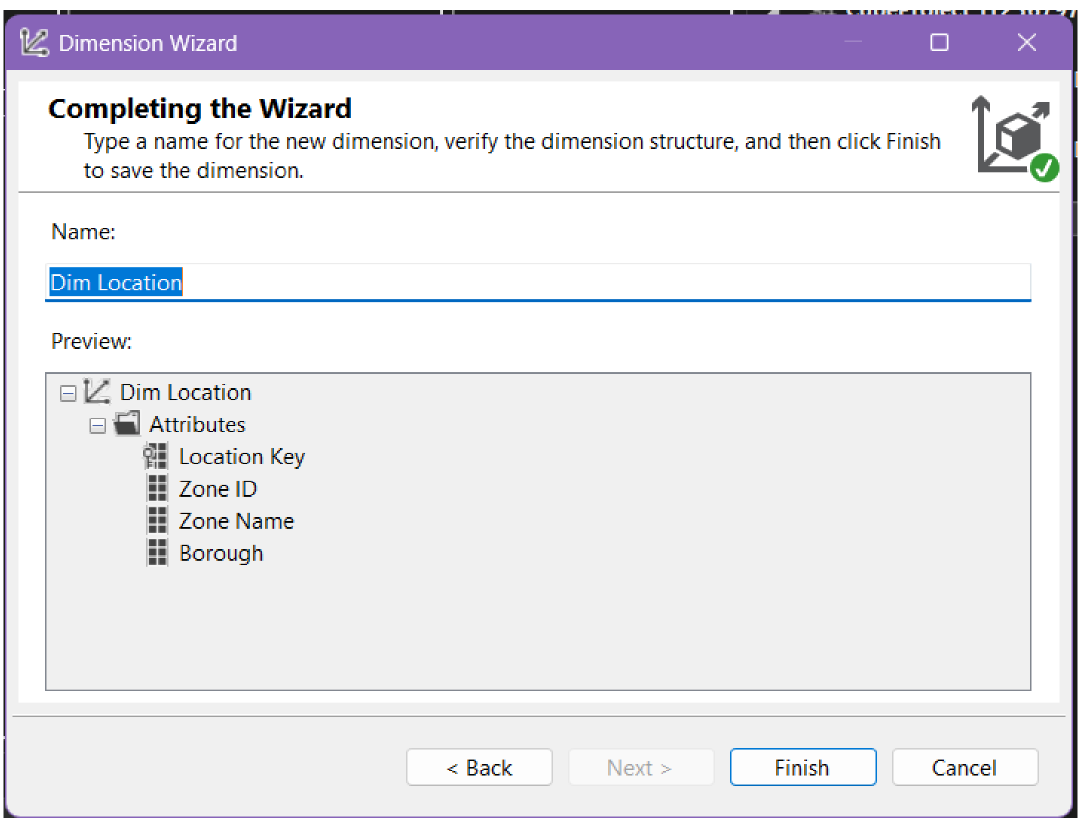

DimPaymentType
PaymentTypeKey
PaymentID
PaymentName
FactTrip
FactTripKey
TripID
PickupDateKey
DropoffDateKey
PickupLocationKey
DropoffLocationKey
PaymentTypeKey
RateCodeKey
VendorKey
StoreAndFwdFlag
PassengerCount
TripDistance
FareAmount
DimLocation
LocationKey
ZoneID
ZoneName
Borough
StartDate
EndDate
IsCurrent
DimRateCode
RateCodeKey
RateCodeID
RateCodeName
DimDate
DateKey
FullDate
DayNumber
MonthNumber
MonthName
QuarterNumber
YearNumber
DayOfWeekNumber
DayOfWeekName
DimVendor
VendorKey
VendorID
VendorName

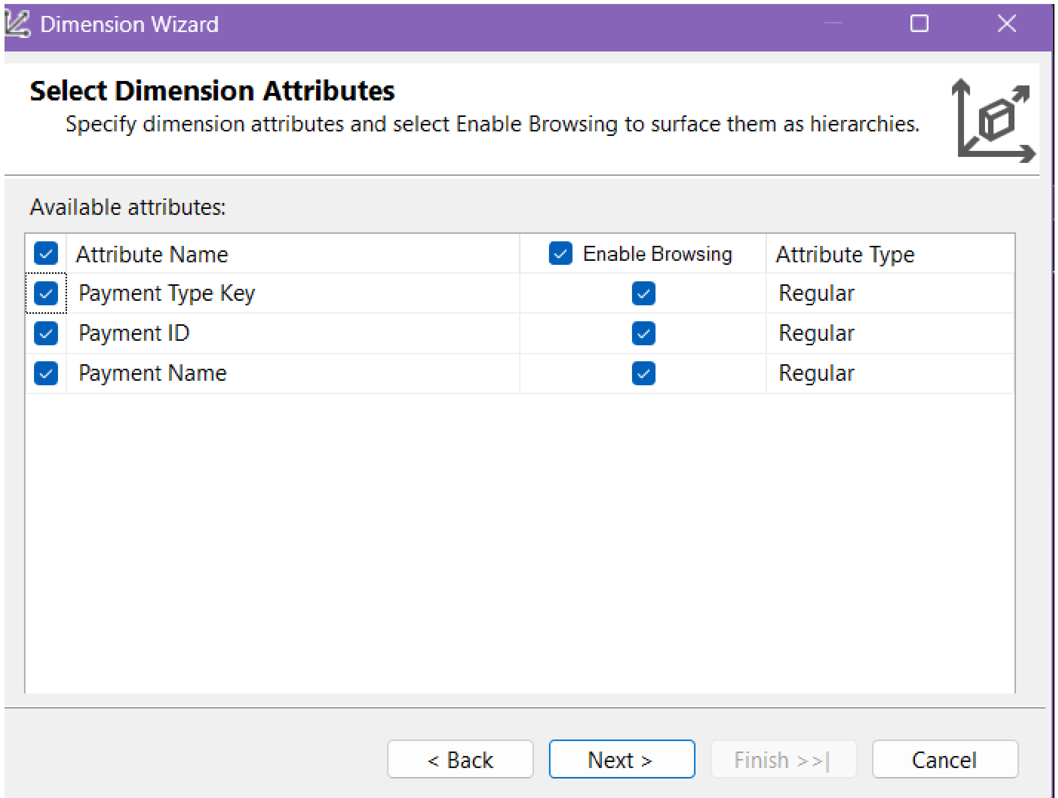
2.3.Create Dimensions.

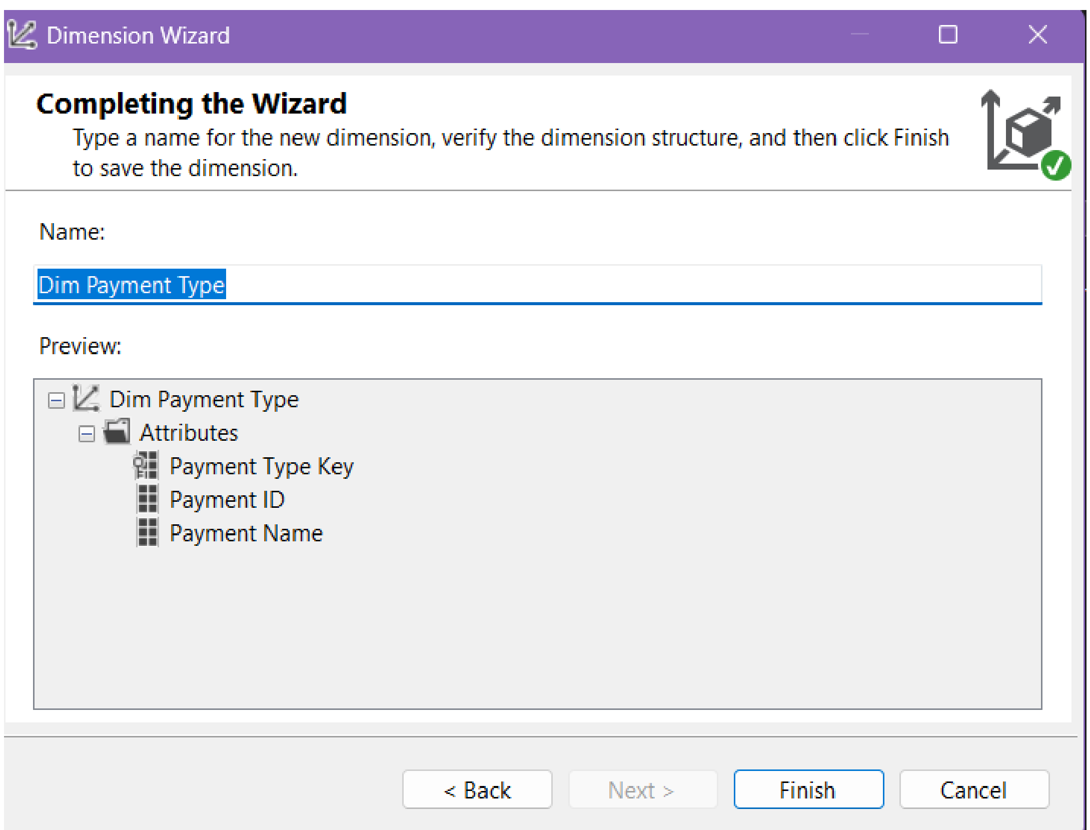


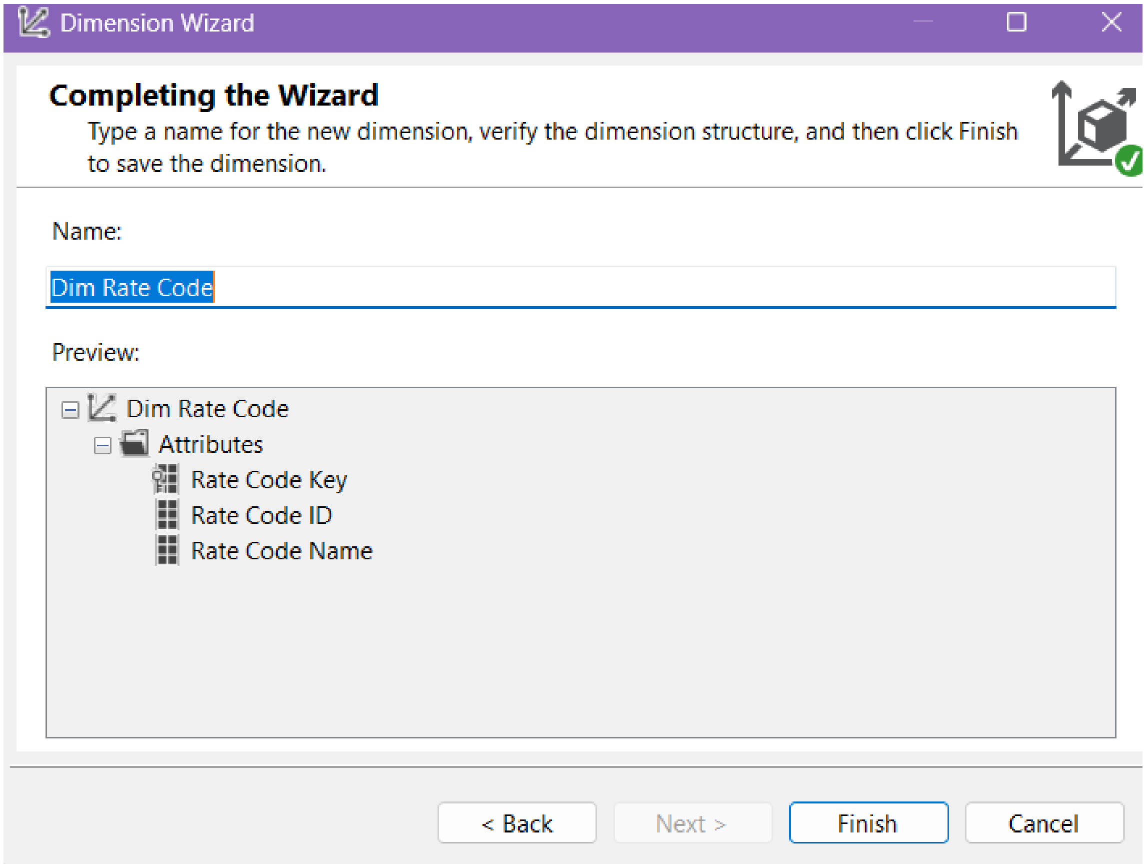


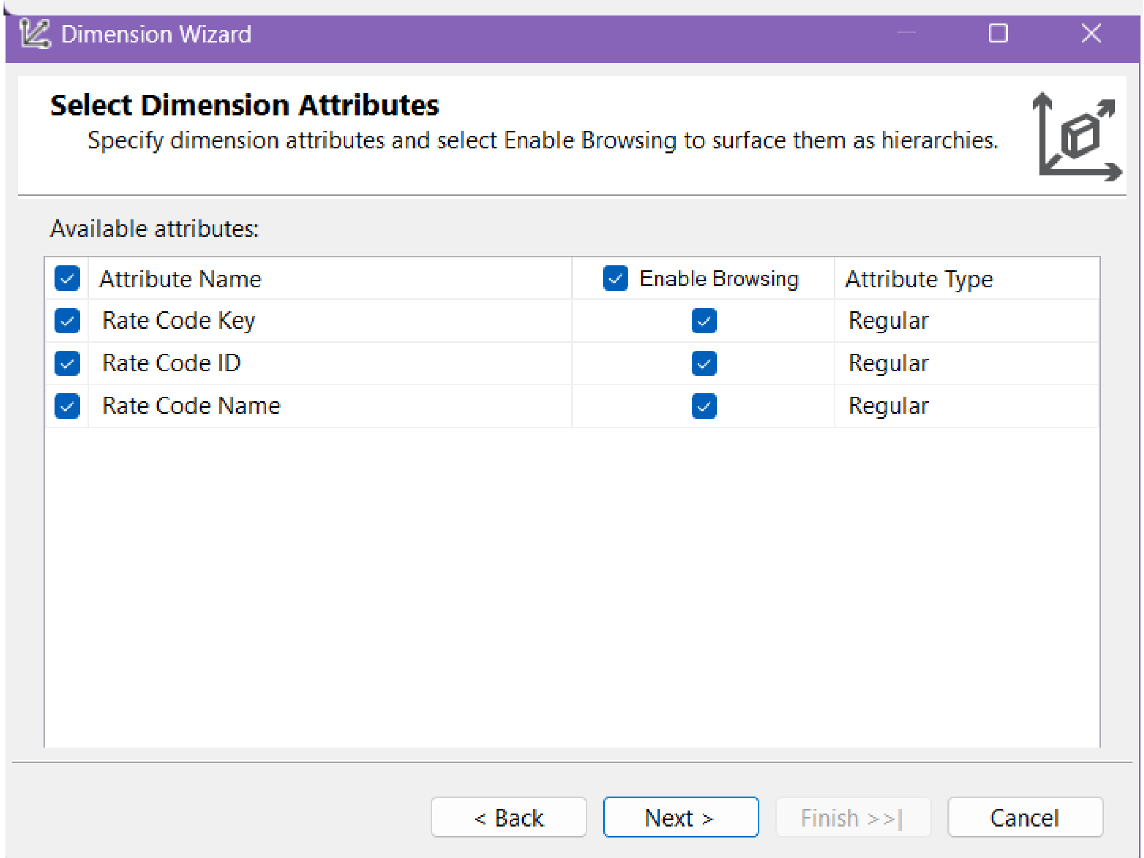

Dimension Wizard
Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.
Available attributes:
Attribute Name
Enable Browsing
Attribute Type
Location Key
Regular
Zone ID
Regular
Zone Name
Regular
Borough
Regular
Start Date
Regular
End Date
Regular
Is Current
Regular
< Back
Next >
Finish >>|
Cancel

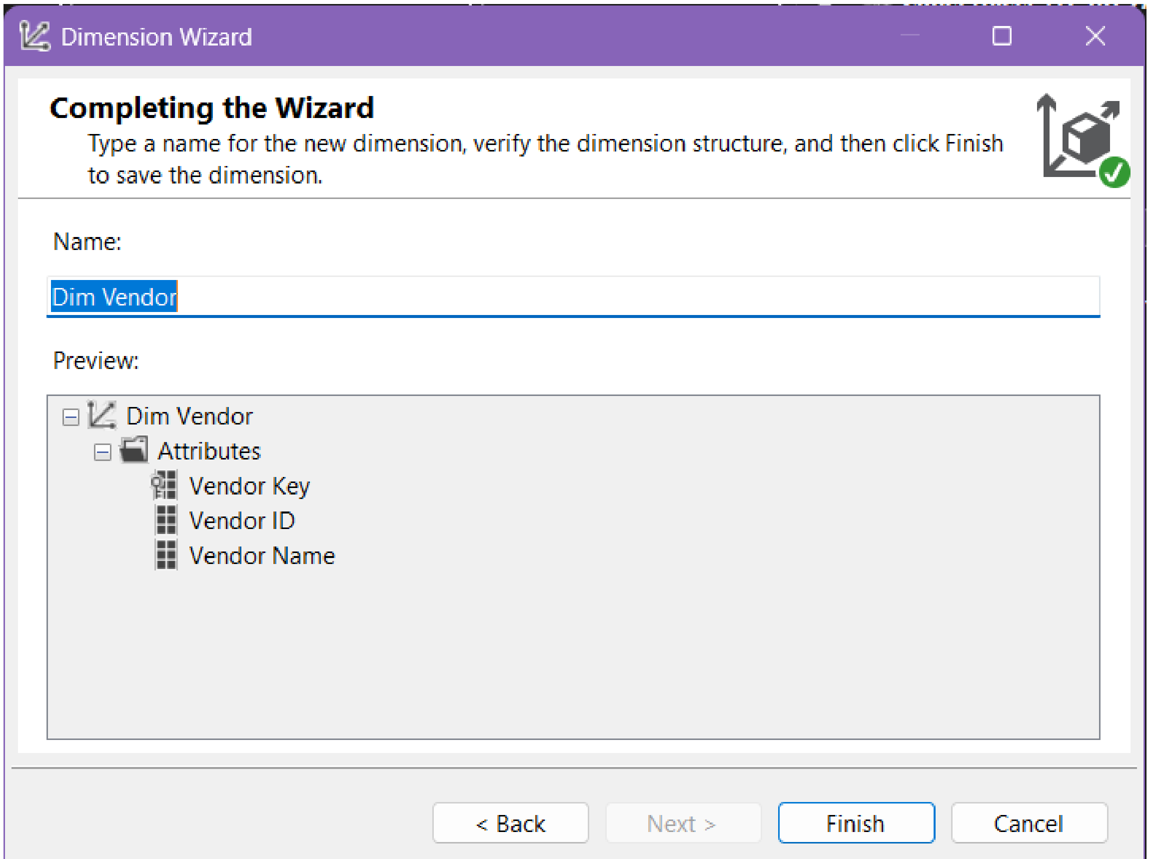

Dimension Wizard
Completing the Wizard
Type a name for the new dimension, verify the dimension structure, and then click Finish to save the dimension.
Name:
Dim Location
Preview:
Dim Location
Attributes
Location Key
Zone ID
Zone Name
Borough
< Back
Next >
Finish
Cancel

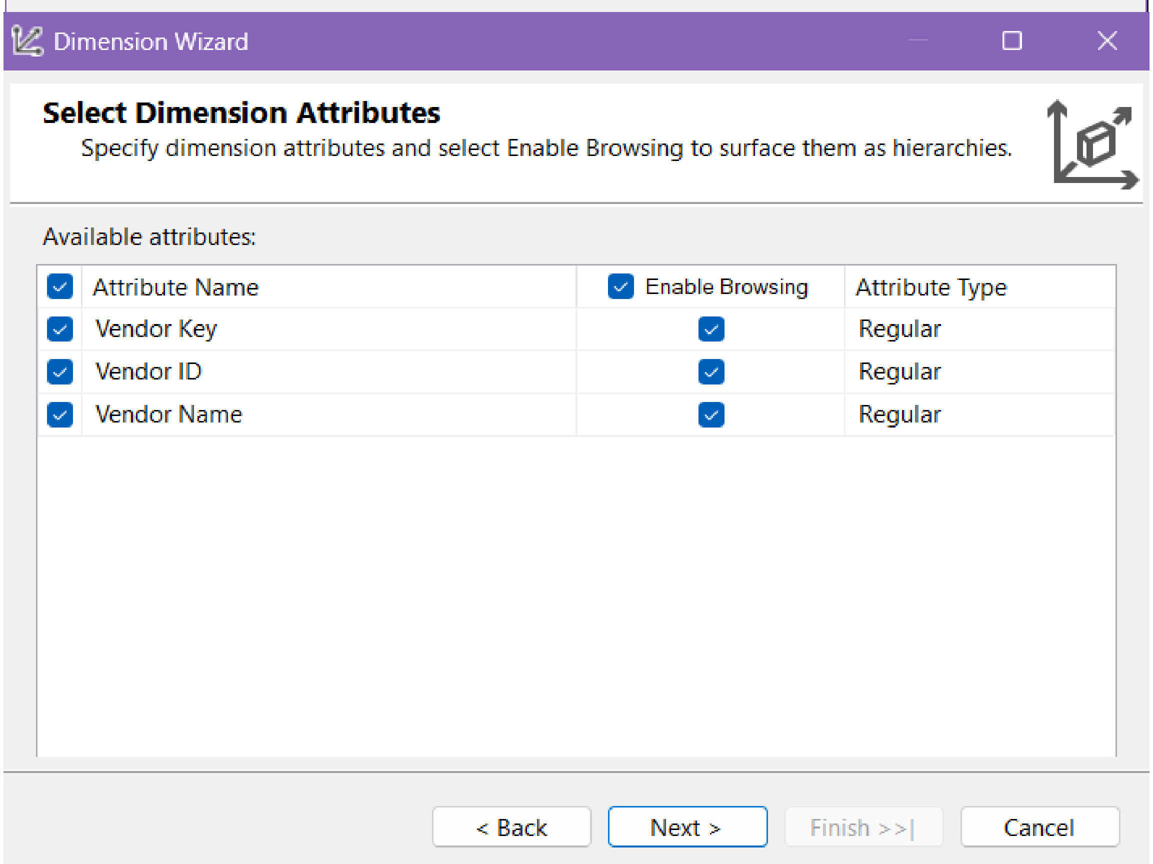

Dimension Wizard
Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.
Available attributes:
Attribute Name
Enable Browsing
Attribute Type
Payment Type Key
Regular
Payment ID
Regular
Payment Name
Regular
< Back
Next >
Finish >>|
Cancel


Dimension Wizard
Completing the Wizard
Type a name for the new dimension, verify the dimension structure, and then click Finish to save the dimension.
Name:
Dim Payment Type
Preview:
Dim Payment Type
Attributes
Payment Type Key
Payment ID
Payment Name
< Back
Next >
Finish
Cancel


Dimension Wizard
Completing the Wizard
Type a name for the new dimension, verify the dimension structure, and then click Finish to save the dimension.
Name:
Dim Rate Code
Preview:
Dim Rate Code
Attributes
Rate Code Key
Rate Code ID
Rate Code Name
< Back
Next >
Finish
Cancel


Dimension Wizard
Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.
Available attributes:
Attribute Name
Enable Browsing
Attribute Type
Rate Code Key
Regular
Rate Code ID
Regular
Rate Code Name
Regular
< Back
Next >
Finish >>|
Cancel


Dimension Wizard
Completing the Wizard
Type a name for the new dimension, verify the dimension structure, and then click Finish to save the dimension.
Name:
Dim Vendor
Preview:
Dim Vendor
Attributes
Vendor Key
Vendor ID
Vendor Name
< Back
Next >
Finish
Cancel

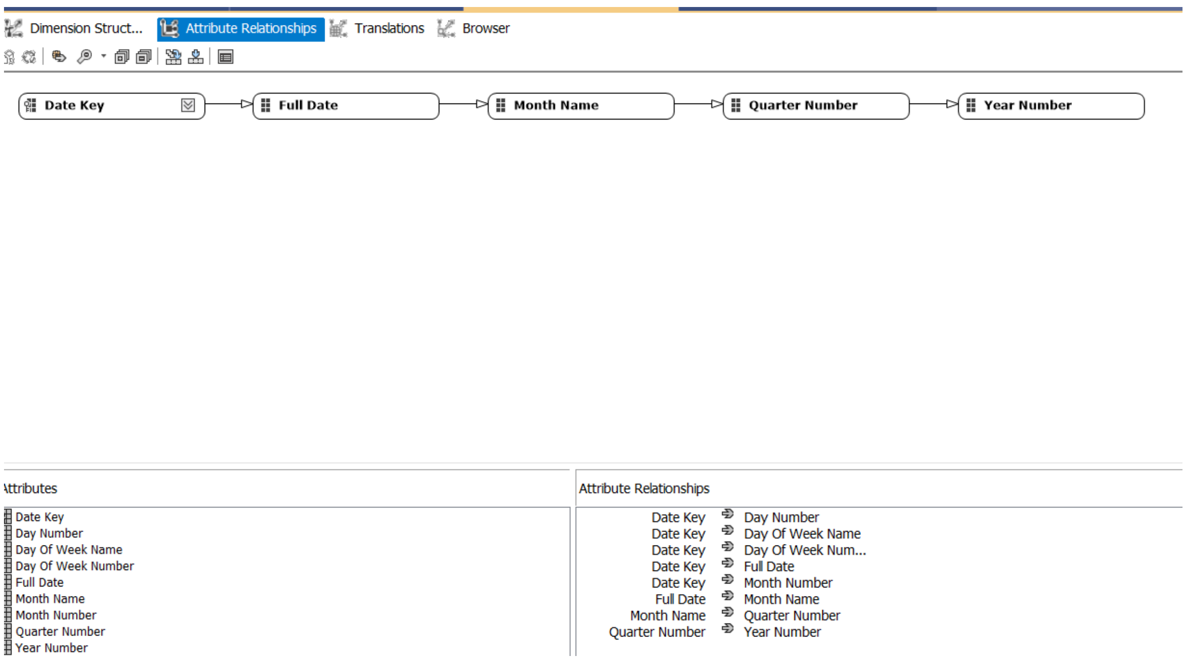

Dimension Wizard
Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.
Available attributes:
Attribute Name
Enable Browsing
Attribute Type
Vendor Key
Regular
Vendor ID
Regular
Vendor Name
Regular
< Back
Next >
Finish >>|
Cancel

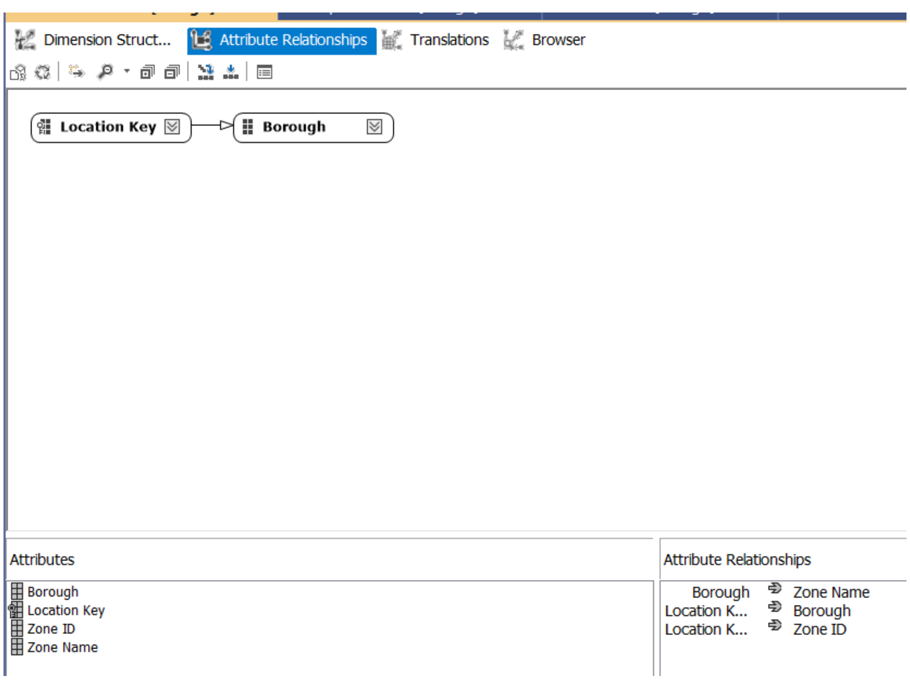
2.4.Create Hierarchies

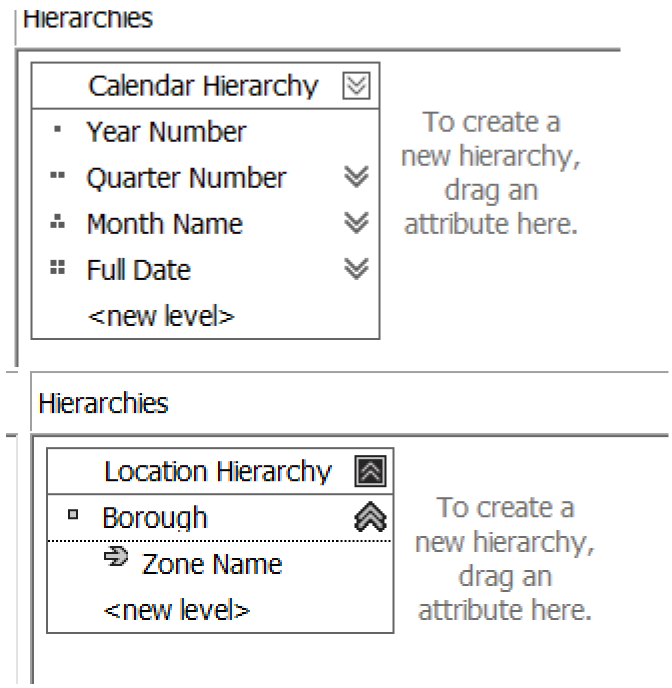
User-defined hierarchies were created to improve cube navigation and support drill-down and roll-up analysis.

Calender Hierarchy – Year, Quarter, Month, FullDate

Location Hierarchy – Borough, Zone Name

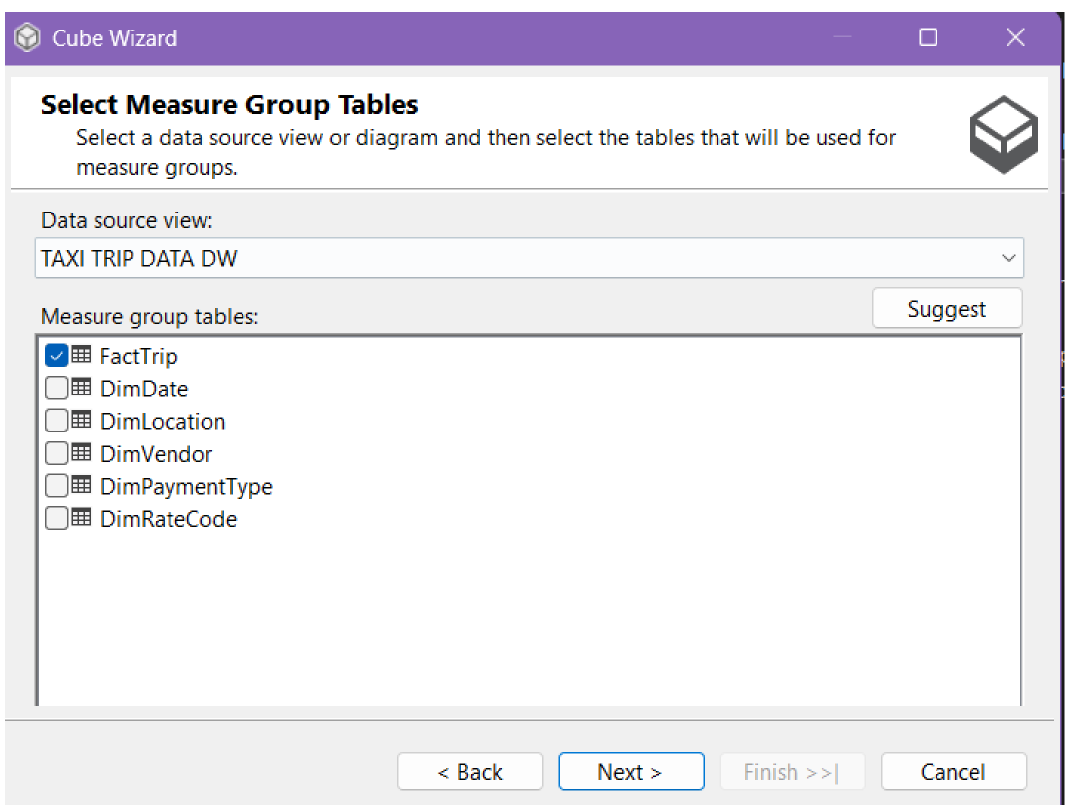


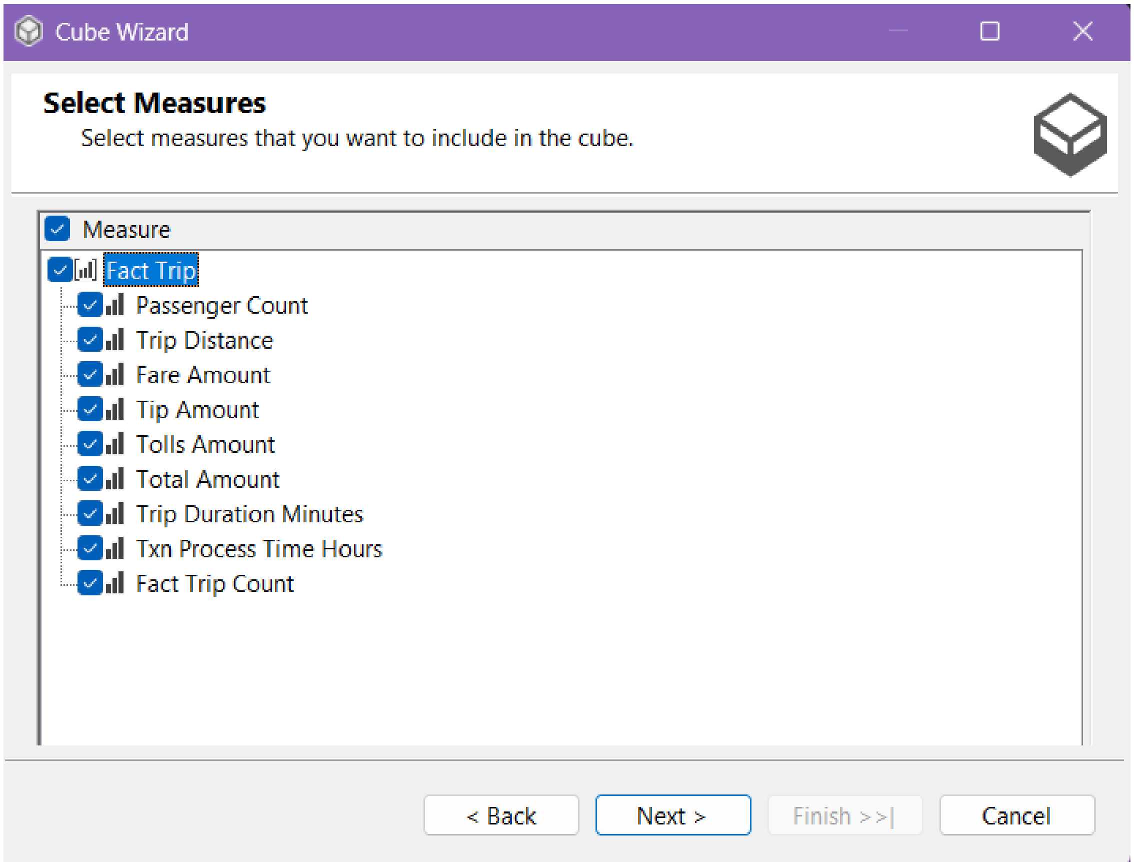


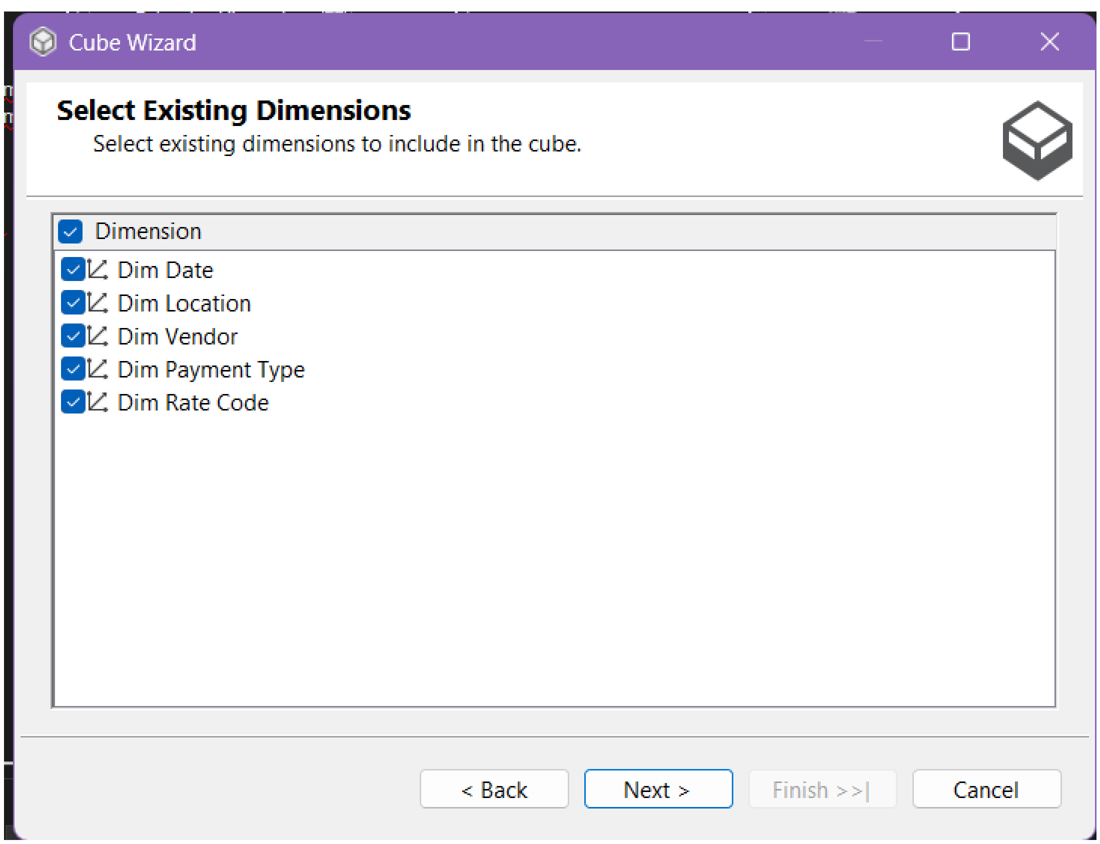


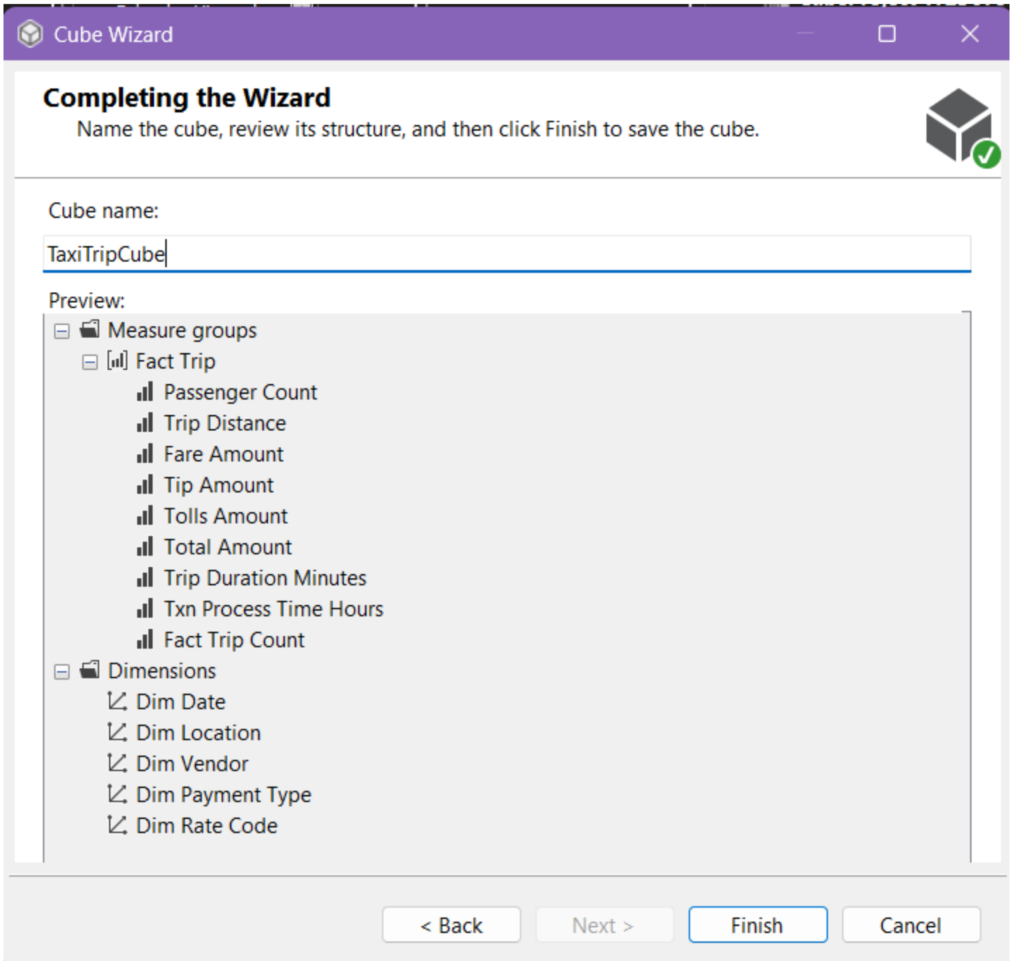
2.5. Create the Cube.

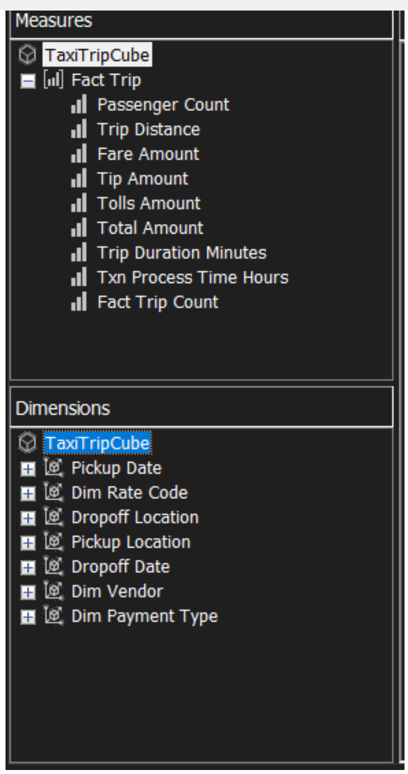
A multidimensional cube named TaxiTripCube was created using FactTrip as the measure group table.




Cube Wizard
Select Measures
Select measures that you want to include in the cube.
Measure
Fact Trip
Passenger Count
Trip Distance
Fare Amount
Tip Amount
Tolls Amount
Total Amount
Trip Duration Minutes
Txn Process Time Hours
Fact Trip Count
< Back
Next >
Finish >>|
Cancel

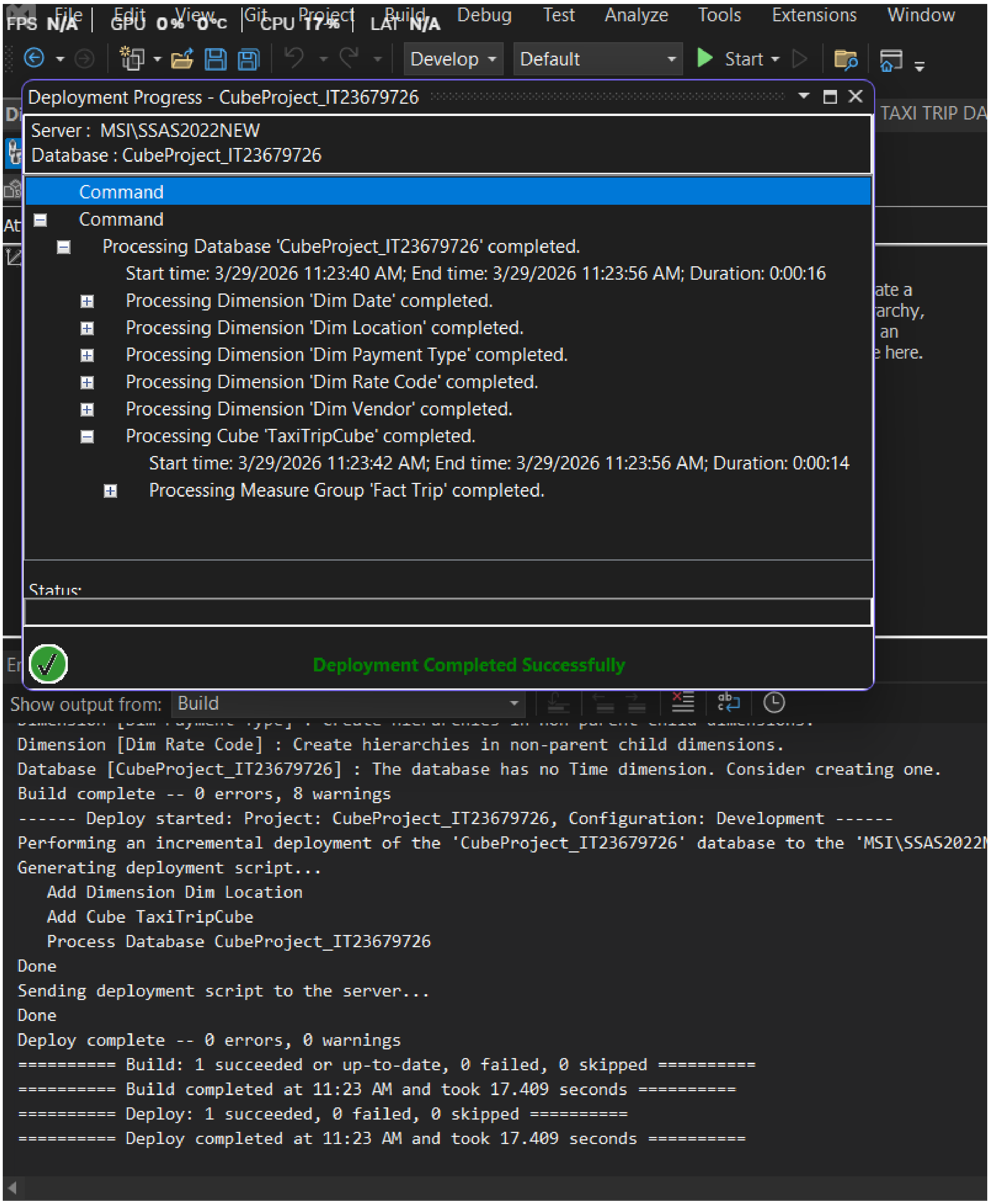

Cube Wizard
Select Existing Dimensions
Select existing dimensions to include in the cube.
Dimension
Dim Date
Dim Location
Dim Vendor
Dim Payment Type
Dim Rate Code
< Back
Next >
Finish >>|
Cancel


Cube Wizard
Completing the Wizard
Name the cube, review its structure, and then click Finish to save the cube.
Cube name:
TaxiTripCube
Preview:
Measure groups
Fact Trip
Passenger Count
Trip Distance
Fare Amount
Tip Amount
Tolls Amount
Total Amount
Trip Duration Minutes
Txn Process Time Hours
Fact Trip Count
Dimensions
Dim Date
Dim Location
Dim Vendor
Dim Payment Type
Dim Rate Code
< Back
Next >
Finish
Cancel

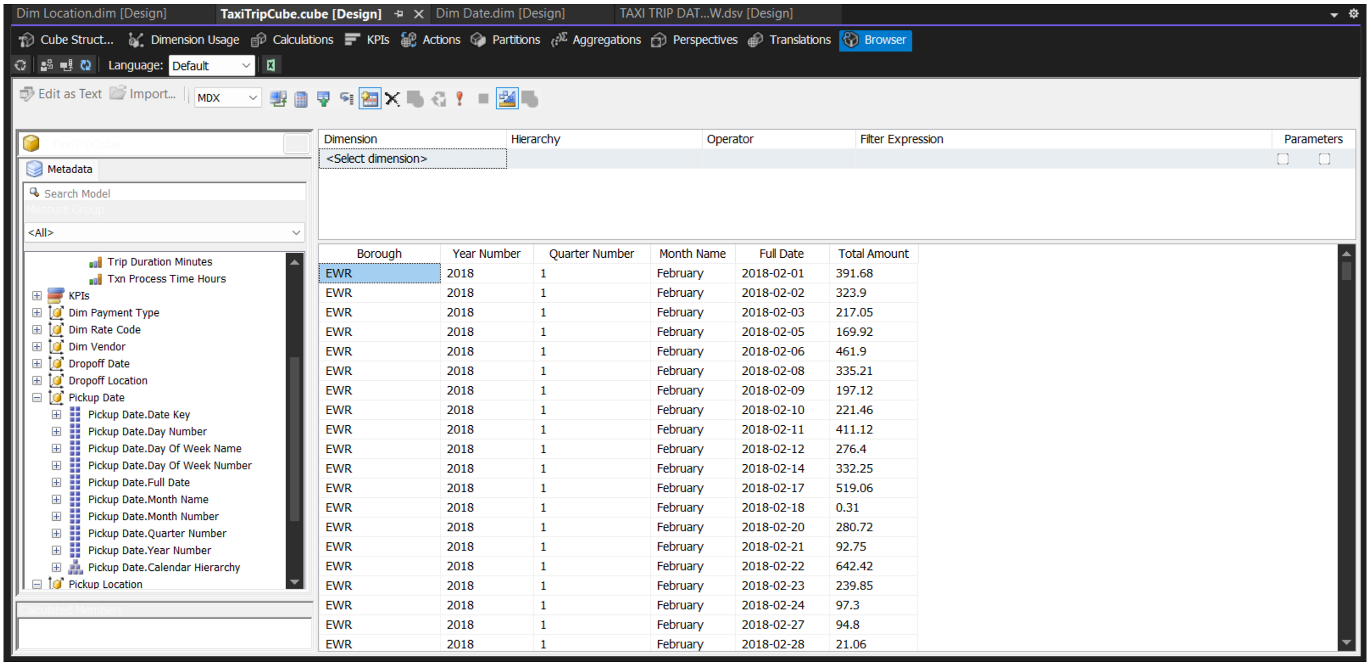

Measures
TaxiTripCube
Fact Trip
Passenger Count
Trip Distance
Fare Amount
Tip Amount
Tolls Amount
Total Amount
Trip Duration Minutes
Txn Process Time Hours
Fact Trip Count
Dimensions
TaxiTripCube
Pickup Date
Dim Rate Code
Dropoff Location
Pickup Location
Dropoff Date
Dim Vendor
Dim Payment Type

2.6.Deploy and Process the Cube

The cube was deployed to the Analysis Services server and processed successfully.

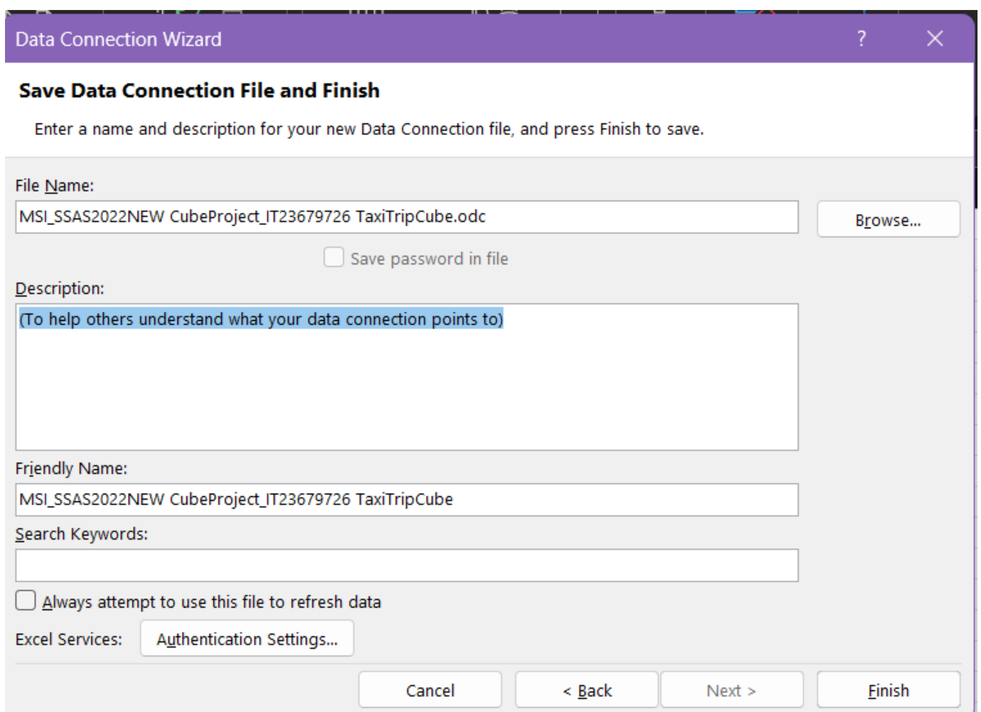


2.7. Validate the Cube in Browser



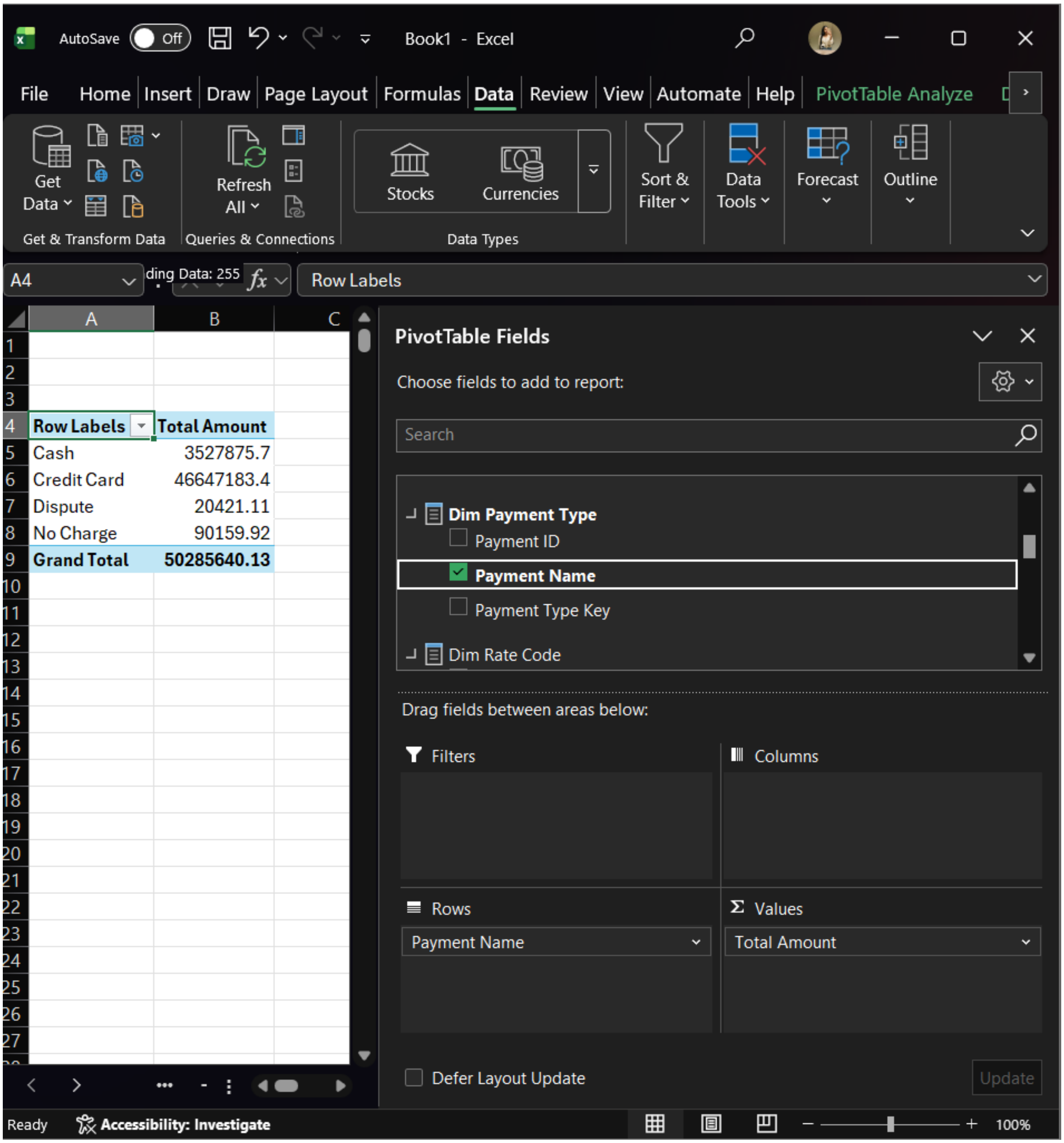
3. Excel Connection and OLAP Operations

The deployed SSAS cube was connected to Microsoft Excel using the Analysis Services connection option.



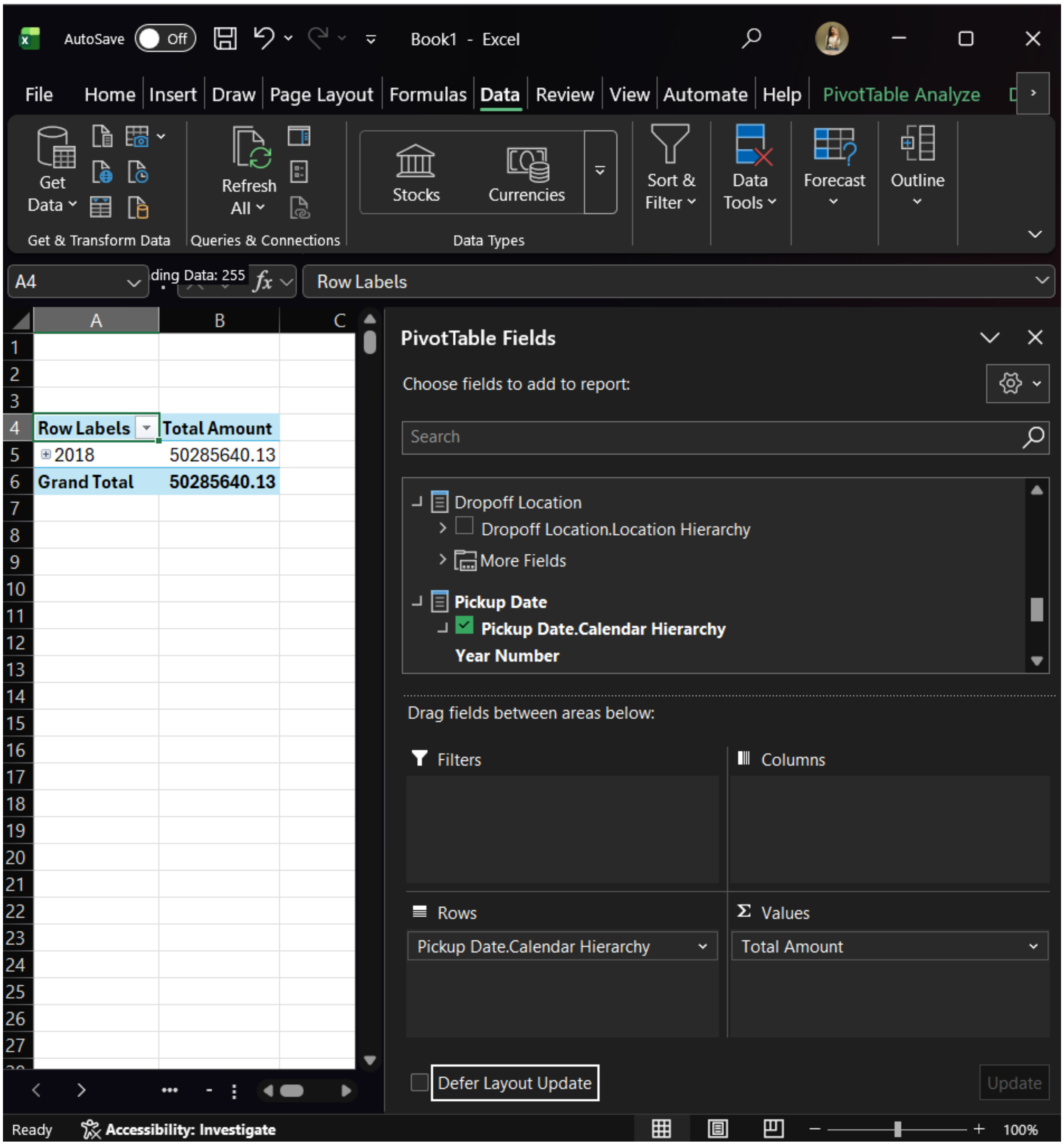
3.1. Initial PivotTable

A basic PivotTable was created to confirm that Excel was successfully connected to the cube. In this report, Total Amount was used as the measure and Payment Name was used as the row field.



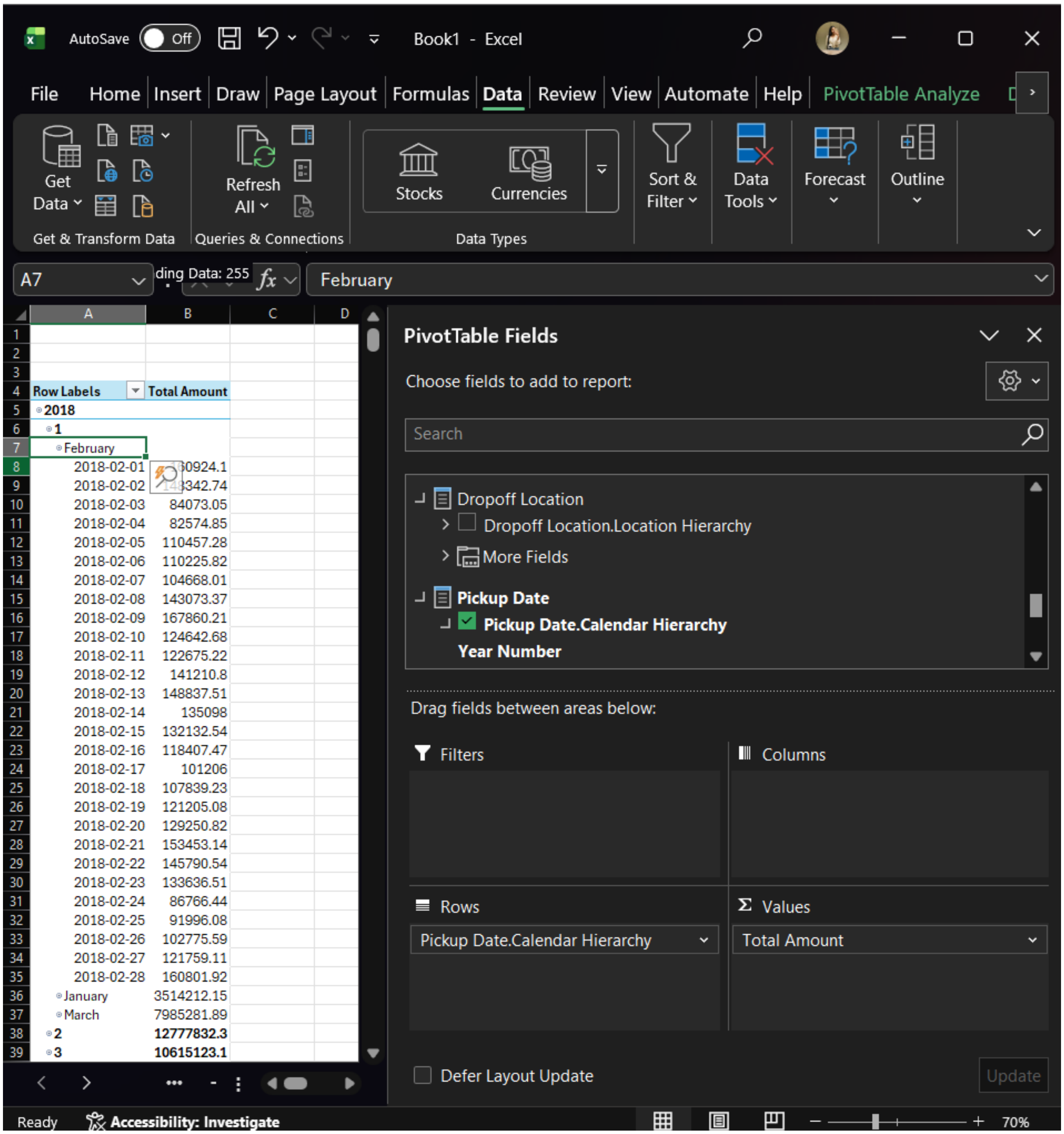
3.2.Roll-up

The roll-up operation was demonstrated using the Calendar Hierarchy in the Pickup Date dimension. Data was summarized at the Year level to provide a higher-level aggregated view of Total Amount.



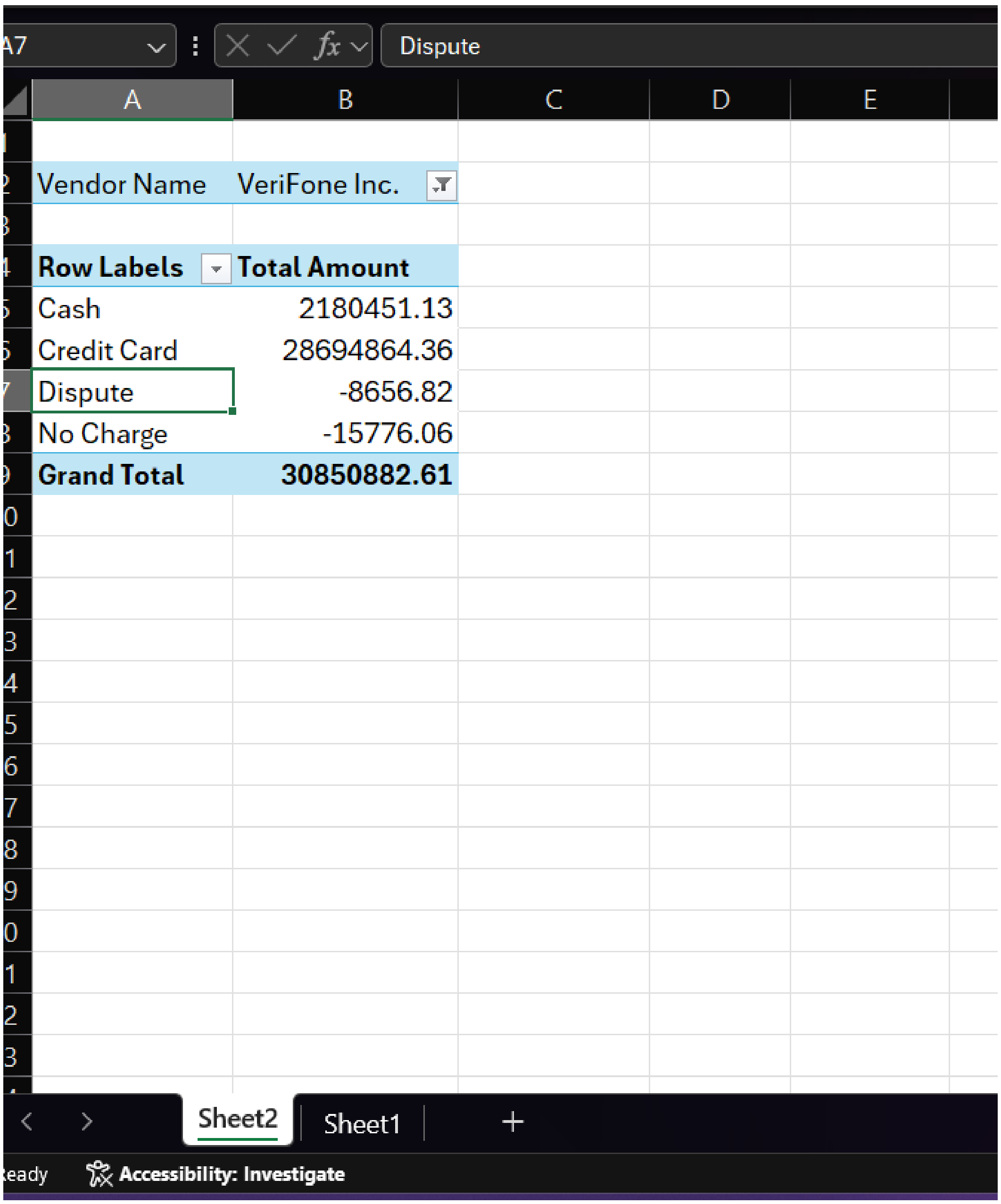
3.3. Drill-down

The drill-down operation was demonstrated by expanding the Calendar Hierarchy from Year to lower levels such as Month and Date. This provided a more detailed view of the revenue data over time.



3.4. Slice

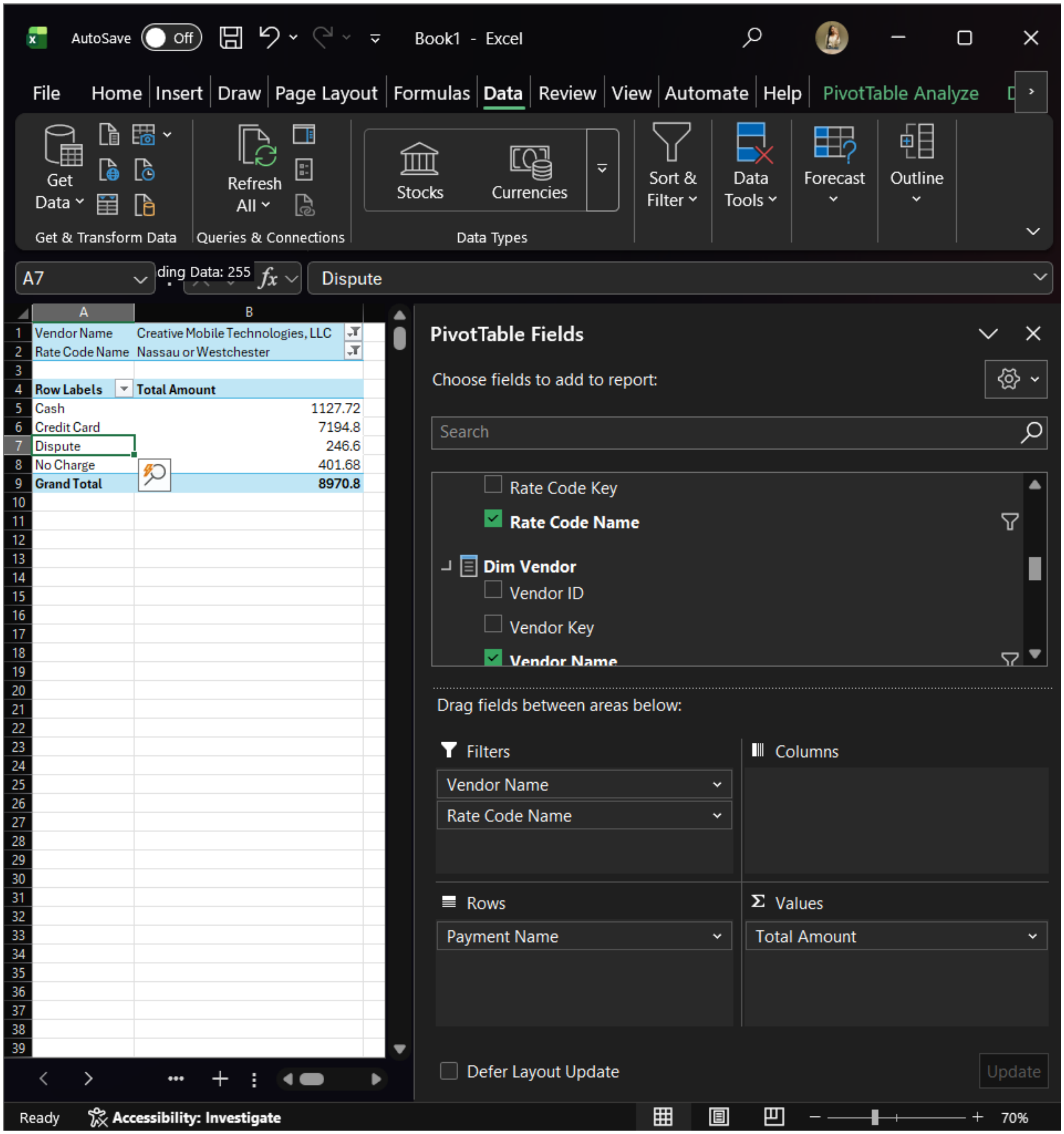
The slice operation was performed by applying a single filter to the PivotTable. For example, the report was filtered by one Vendor Name, allowing the data to be viewed for only one selected vendor.



Vendor Name	VeriFone Inc.
Row Labels	**Total Amount**
Cash	2180451.13
Credit Card	28694864.36
Dispute	-8656.82
No Charge	-15776.06
Grand Total	**30850882.61**

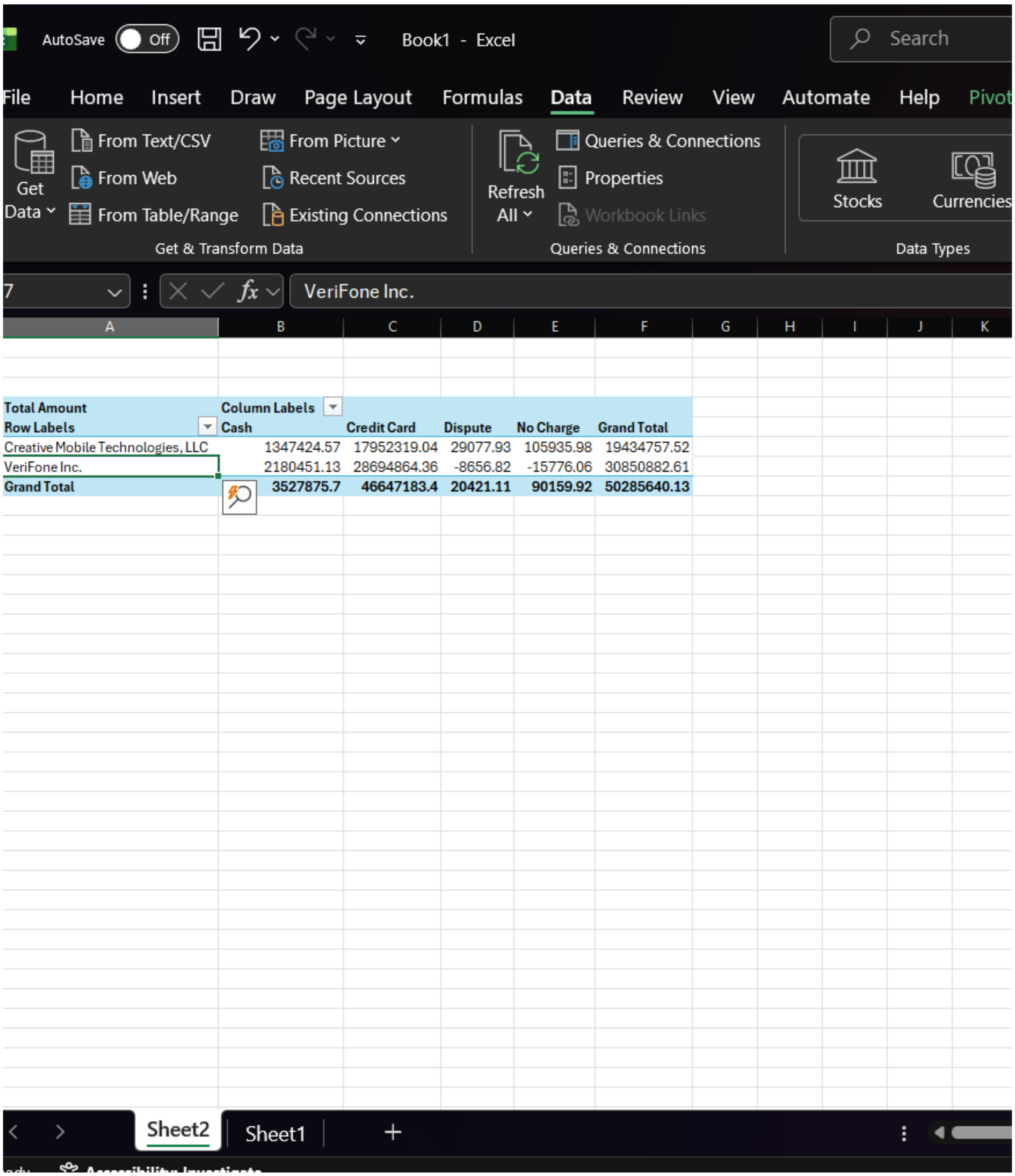
3.5.Dice

The dice operation was demonstrated by applying multiple filters simultaneously. In this case, the PivotTable was filtered using Vendor Name and Rate Code Name, producing a smaller subset of cube data for analysis.



3.6.Pivot

The pivot operation was demonstrated by changing the arrangement of dimensions between the row and column areas of the PivotTable. In the final view, Vendor Name was placed in the rows while Payment Name was placed in the columns, allowing the same data to be viewed from a different analytical perspective.



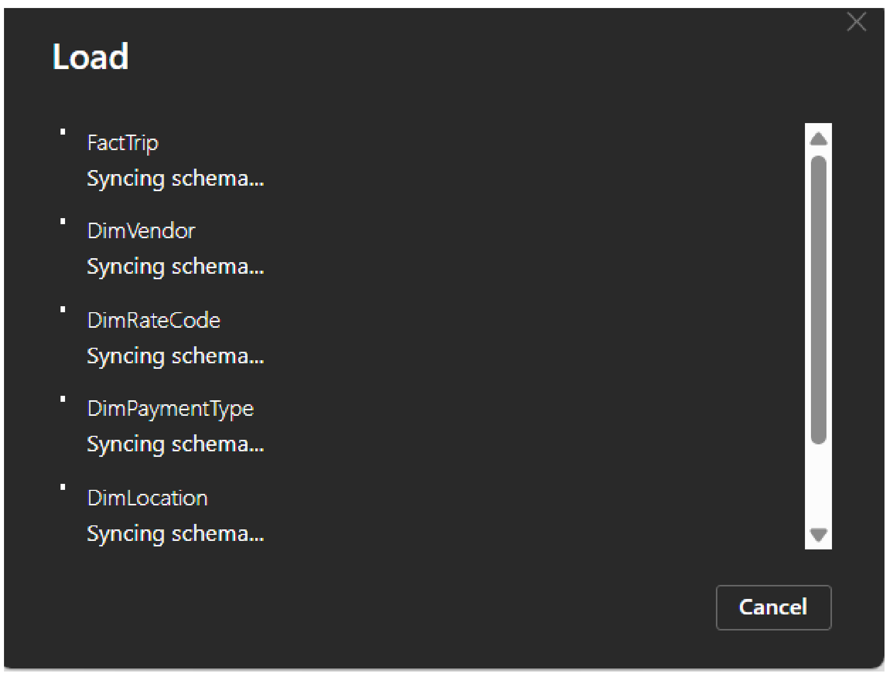
Total Amount	Column Labels				
Row Labels	Cash	Credit Card	Dispute	No Charge	Grand Total
Creative Mobile Technologies, LLC	1347424.57	17952319.04	29077.93	105935.98	19434757.52
VeriFone Inc.	2180451.13	28694864.36	-8656.82	-15776.06	30850882.61
Grand Total	3527875.7	46647183.4	20421.11	90159.92	50285640.13

4. **Power BI Reports.**

4.1.Data Preparation and Modeling

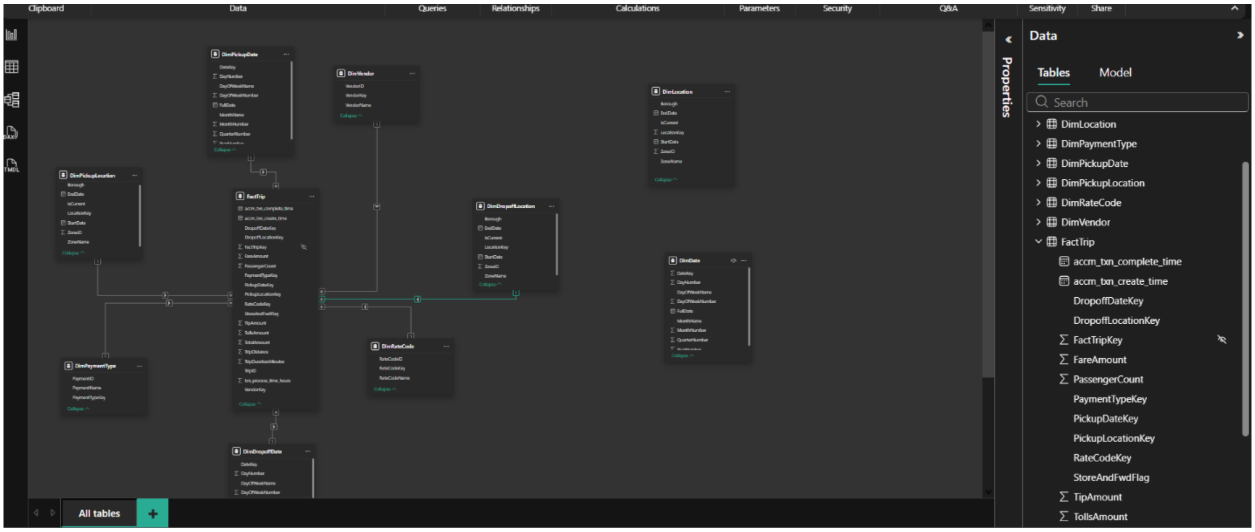
Power BI Desktop was used to create the required reports. The data was imported from the TAXI_TRIP_DATA_DW SQL Server database. The tables loaded were:

- FactTrip
- DimDate
- DimLocation
- DimVendor
- DimPaymentType
- DimRateCode



To support pickup and drop-off reporting clearly, duplicate dimension tables were created in Power BI:

- DimPickupDate
- DimDropoffDate
- DimPickupLocation
- DimDropoffLocation



Relationships were configured between the fact table and these dimensions. In addition, DAX measures were created for reporting purposes, including:

- Total Revenue
- Total Trips
- Total Distance
- Total Passengers
- Average Fare

```
Total Trips = COUNTROWS(FactTrip)
```

```
Total Revenue = SUM(FactTrip[TotalAmount])
```

```
Total Passengers = SUM(FactTrip[PassengerCount])
```

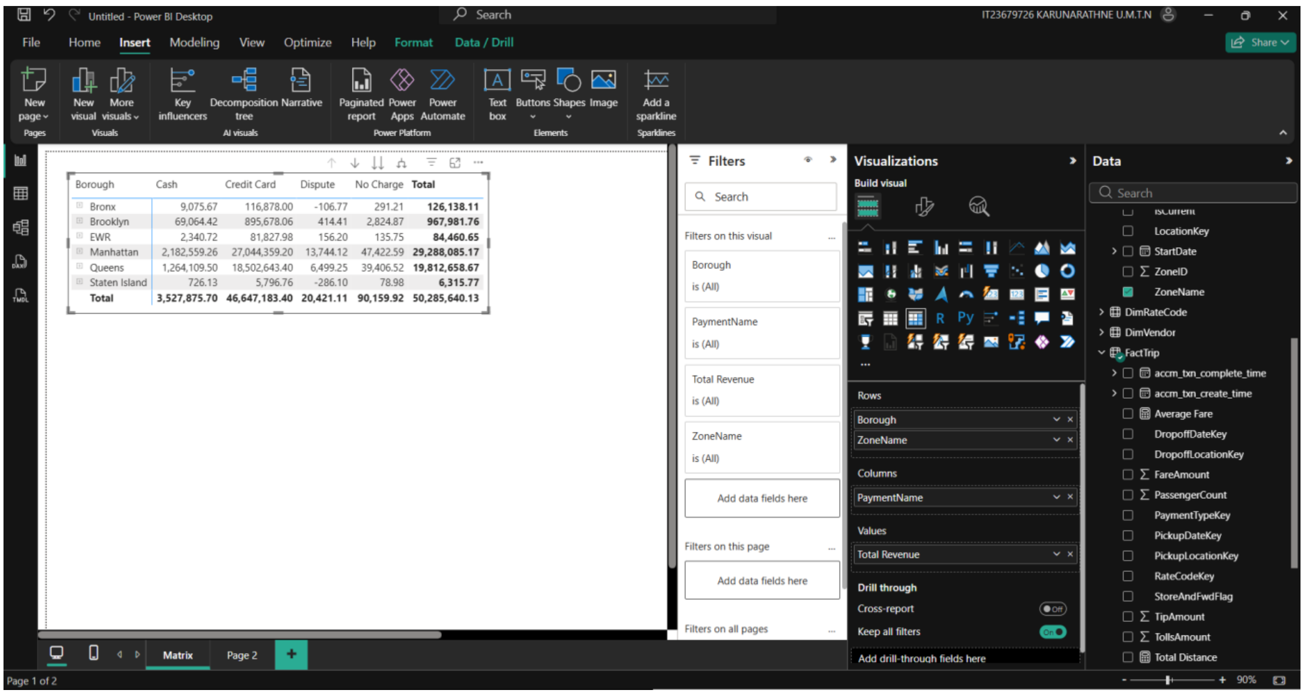
```
Total Distance = SUM(FactTrip[TripDistance])
```

```
Average Fare = AVERAGE(FactTrip[FareAmount])
```

4.2. Report 1 – Matrix Visual

The report used Borough and ZoneName from the Pickup Location dimension as row groups, PaymentName as the column group, and Total Revenue as the measure.

This report provides a compact analytical summary of revenue distribution across locations and payment methods.

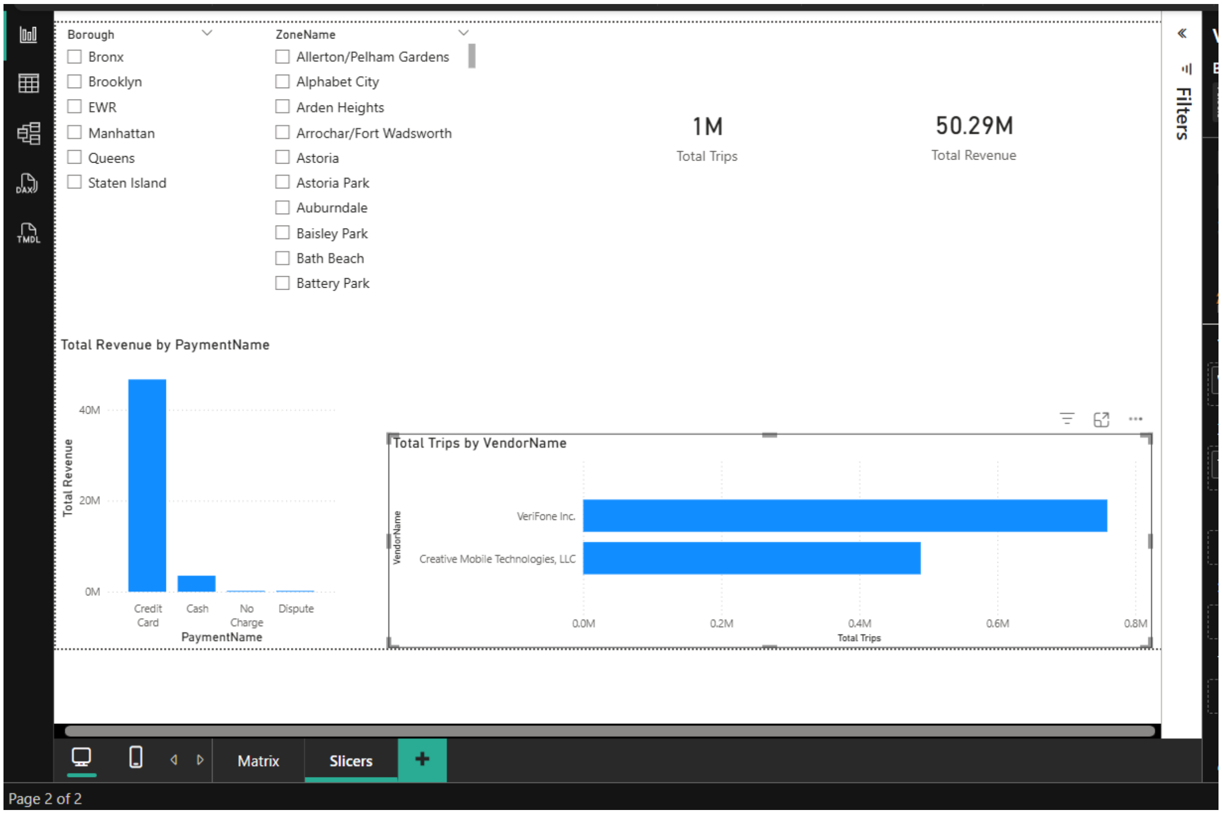


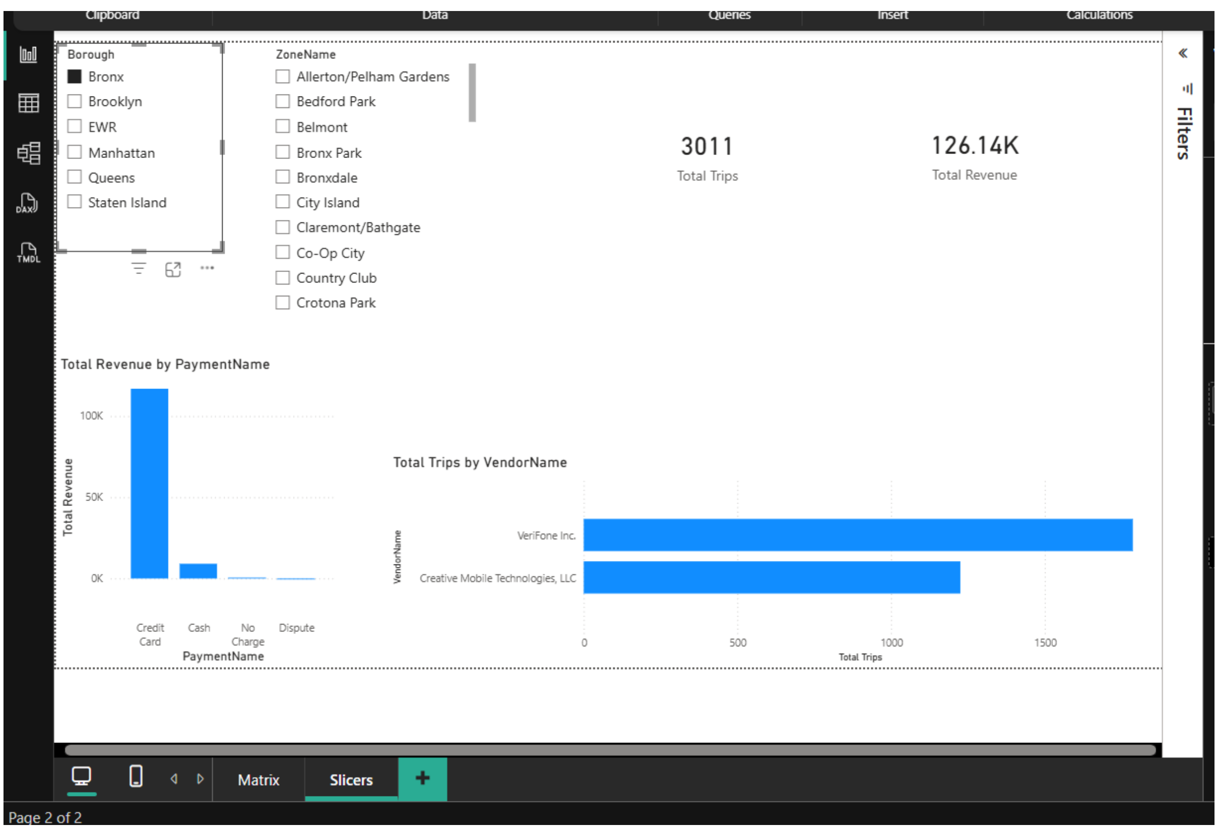
4.3. Multiple Slicers with Cascading Filtering

The first slicer used Borough and the second slicer used ZoneName from the Pickup Location dimension. When a Borough was selected, the ZoneName slicer displayed only the zones related to that borough, demonstrating cascading filtering.

The report also included:

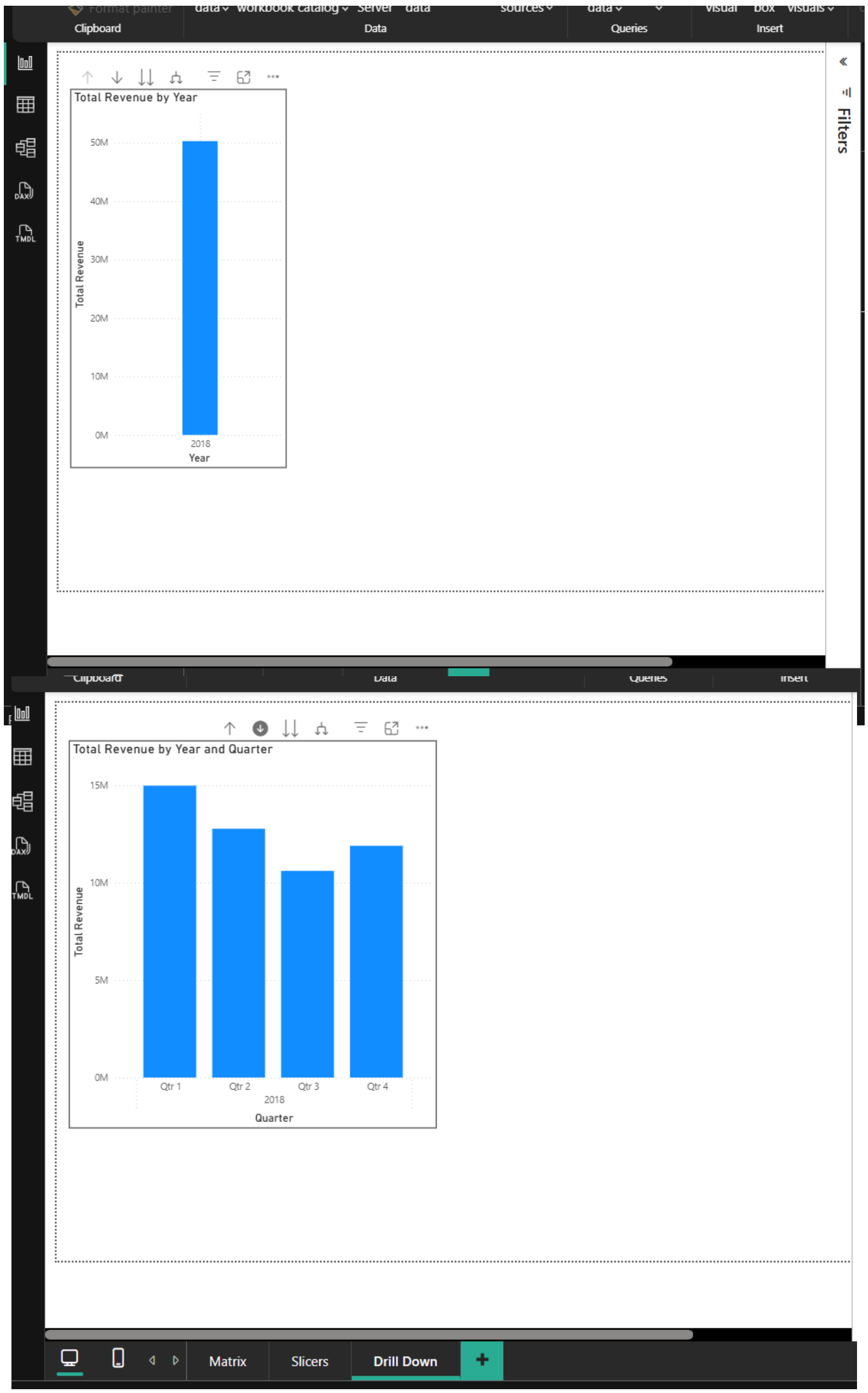
- a column chart for Total Revenue by Payment Type
- a bar chart for Total Trips by Vendor
- card visuals for Total Trips and Total Revenue

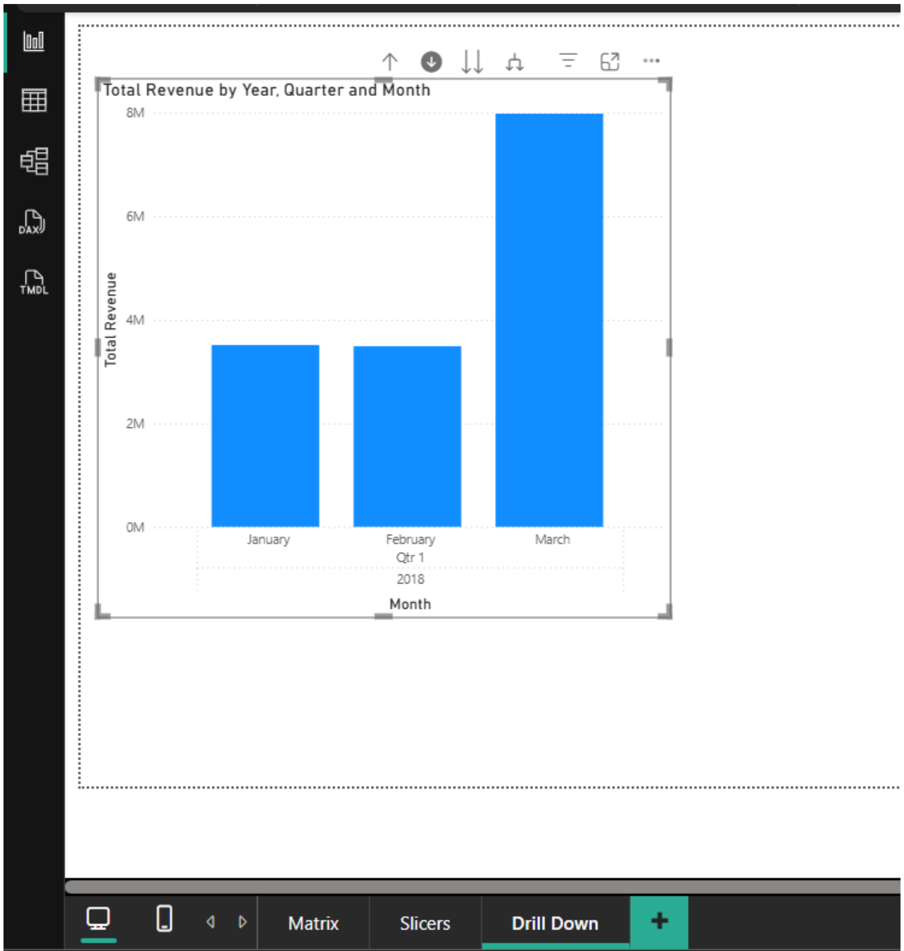

Borough
Bronx
Brooklyn
EWR
Manhattan
Queens
Staten Island
ZoneName
Allerton/Pelham Gardens
Alphabet City
Arden Heights
Arrochar/Fort Wadsworth
Astoria
Astoria Park
Auburndale
Baisley Park
Bath Beach
Battery Park
1M
Total Trips
50.29M
Total Revenue
Filters
Total Revenue by PaymentName
40M
20M
0M
Total Revenue
Credit Card
Cash
No Charge
Dispute
PaymentName
Total Trips by VendorName
VeriFone Inc.
Creative Mobile Technologies, LLC
VendorName
0.0M
0.2M
0.4M
0.6M
0.8M
Total Trips
Matrix
Slicers
Page 2 of 2

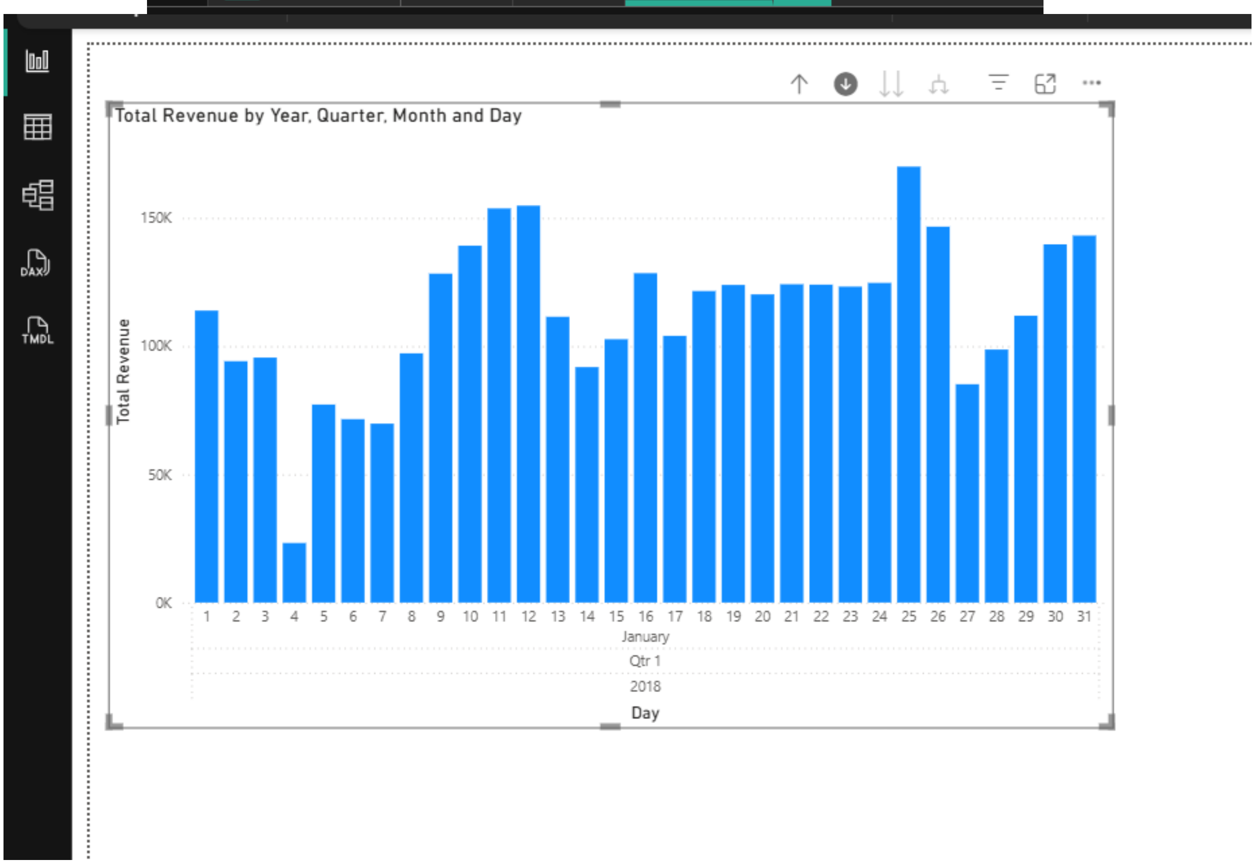

Clipboard
Data
Queries
Insert
Calculations
Borough
Bronx
Brooklyn
EWR
Manhattan
Queens
Staten Island
ZoneName
Allerton/Pelham Gardens
Bedford Park
Belmont
Bronx Park
Bronxdale
City Island
Claremont/Bathgate
Co-Op City
Country Club
Crotona Park
3011
Total Trips
126.14K
Total Revenue
Filters
Total Revenue by PaymentName
100K
50K
0K
Total Revenue
Credit Card
Cash
No Charge
Dispute
PaymentName
Total Trips by VendorName
VeriFone Inc.
Creative Mobile Technologies, LLC
VendorName
0
500
1000
1500
Total Trips
Matrix
Slicers
Page 2 of 2

4.4. Report 3 – Drill-Down Report

A drill-down report was created in Power BI using the Date Hierarchy available in the Pickup Date dimension. A clustered column chart was used to display Total Revenue across time. The visual initially displayed yearly values and was then drilled down to lower levels such as quarter, month, and day.




Total Revenue by Year, Quarter and Month
8M
6M
4M
2M
0M
Total Revenue
January
February
March
Qtr 1
2018
Month
Matrix
Slicers
Drill Down

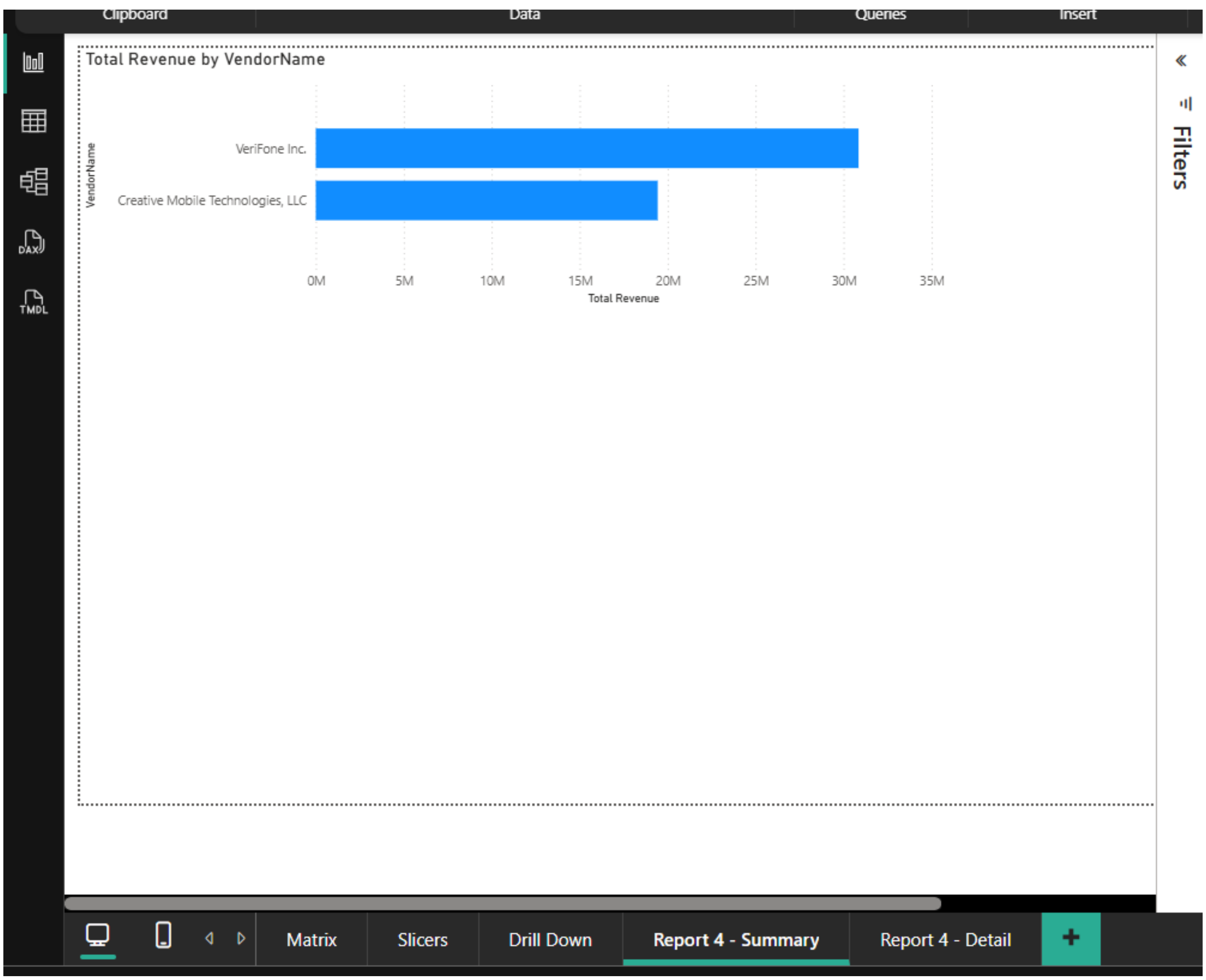

Total Revenue by Year, Quarter, Month and Day
150K
100K
50K
0K
Total Revenue
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
January
Qtr 1
2018
Day

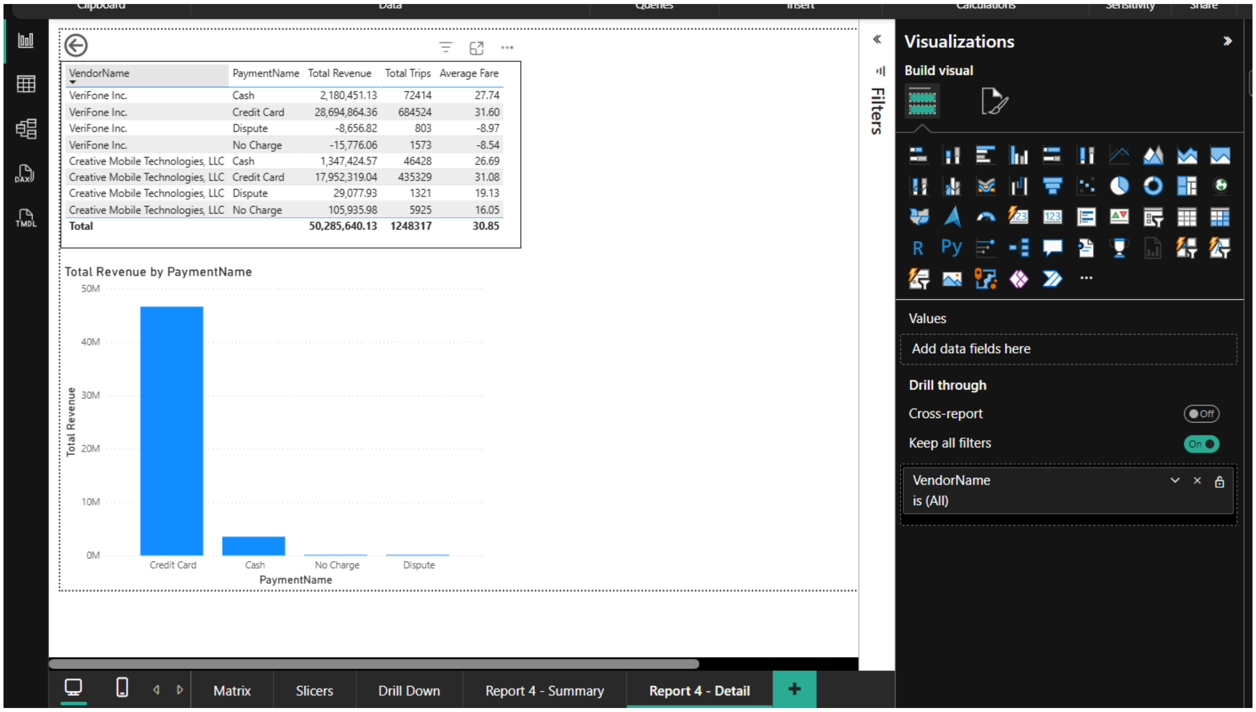
4.5.Report 4 – Drill-Through Report

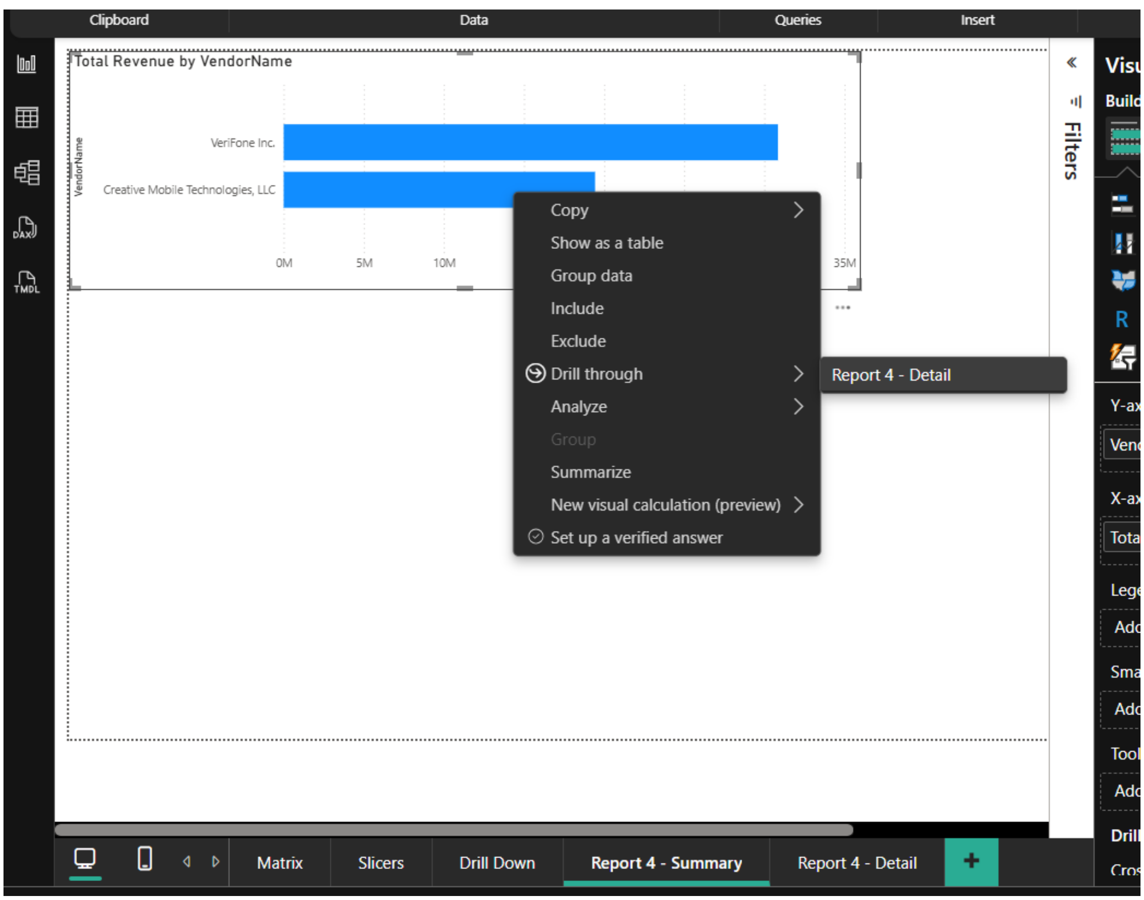
A drill-through report was created to enable navigation from a summary page to a detailed page. The summary page displayed Total Revenue by Vendor using a bar chart. A separate detail page was configured with VendorName as the drill-through field.

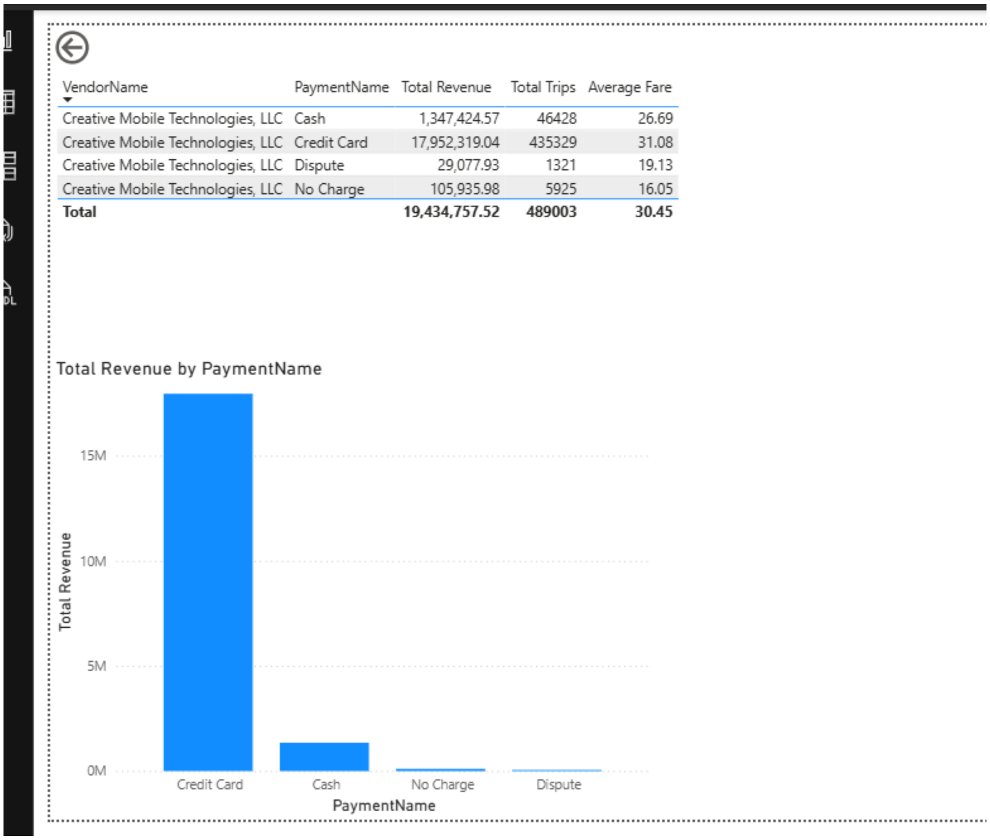
When a user right-clicked a vendor in the summary report and selected the drill-through option, Power BI opened the detail page filtered to the selected vendor. The detail page included:

- a table visual with VendorName, PaymentName, Total Revenue, Total Trips, and Average Fare
- a chart showing Total Revenue by PaymentName
- a back button for navigation to the summary page




VendorName
PaymentName
Total Revenue
Total Trips
Average Fare
VeriFone Inc. Cash 2,180,451.13 72414 27.74
VeriFone Inc. Credit Card 28,694,864.36 684524 31.60
VeriFone Inc. Dispute -8,656.82 803 -8.97
VeriFone Inc. No Charge -15,776.06 1573 -8.54
Creative Mobile Technologies, LLC Cash 1,347,424.57 46428 26.69
Creative Mobile Technologies, LLC Credit Card 17,952,319.04 435329 31.08
Creative Mobile Technologies, LLC Dispute 29,077.93 1321 19.13
Creative Mobile Technologies, LLC No Charge 105,935.98 5925 16.05
Total 50,285,640.13 1248317 30.85
Total Revenue by PaymentName
50M
40M
30M
20M
10M
0M
Total Revenue
Credit Card
Cash
No Charge
Dispute
PaymentName
Filters
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
VendorName
is (All)
Matrix
Slicers
Drill Down
Report 4 - Summary
Report 4 - Detail


Clipboard
Data
Queries
Insert
Total Revenue by VendorName
VeriFone Inc.
Creative Mobile Technologies, LLC
VendorName
0M
5M
10M
35M
Copy
Show as a table
Group data
Include
Exclude
Drill through
Report 4 - Detail
Analyze
Group
Summarize
New visual calculation (preview)
Set up a verified answer
Filters
Matrix
Slicers
Drill Down
Report 4 - Summary
Report 4 - Detail



4.6. Publish to Power BI Service

After creating the reports in Power BI Desktop, the .pbix file was published to Power BI Service. The report was then opened online to confirm that the published version was accessible and functioning correctly.